



Multi-label feature selection by strongly relevant label gain and label mutual aid

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ABSTRACT

Multi-label feature selection, which addresses the challenge of high dimensionality in multi-label learning, has wide applicability in pattern recognition, machine learning, and related domains. Most existing studies on multi-label feature selection assume that all labels have the same importance with respect to features, however, they overlook the differences between labels and candidate features relative to selected features and the internal influence of the label space. To address this issue, we propose a novel method for multi-label feature selection that accounts for both the strongly relevant label gain and the label mutual aid. Firstly, we advance two new potential relationships between labels and candidate features relative to selected features, and the label discriminant function is introduced. Secondly, the mutual aid information between labels is presented to describe the internal correlation of the label space. Thirdly, the concept of strongly relevant label gain is defined based on the label discriminant function, which allows better exploration of positive correlation between features. Finally, the experimental results on sixteen multi-label benchmark datasets indicate that the proposed method outperforms other compared representative multi-label feature selection methods.

1. Introduction

Traditional single-label learning [1,2] typically focuses on identifying a single label, while real-world data are more complex, with each sample possibly being associated with multiple labels. In the face of these more elaborate multi-label data, traditional single-label learning can no longer meet the demand. Therefore, to enhance the effectiveness of processing multi-label data, the concept of multi-label learning has been introduced. At present, multi-label learning has received widespread attention in numerous fields, including image annotation [3], audio and video classification [4], and text recognition [5], etc. However, the performance of multi-label learning algorithms heavily depends on the quality of available features, and the learning algorithms often suffer from the curse of dimensionality.

Feature selection is a process of reducing the dimensionality of the feature space, which aims to select the most relevant subset of features and remove redundant and irrelevant features. Therefore, multi-label feature selection can effectively address the issue of the curse of dimensionality in multi-label learning. Researchers have proposed several feasible methods for multi-label feature selection in dealing with this issue [6,7]. From the perspective of different selection strategies, the multi-label feature selection methods can be divided into

filter, wrapper and embedded methods [8]. Filter methods aim to reduce computational costs while maintaining classification performance. These methods are independent and do not impact subsequent learning algorithms [9]. Wrapper methods are frequently utilized in combination with subsequent learning algorithms. However, the wrapper methods are accompanied by the problem of over-fitting [6]. Embedded methods have also been combined with subsequent learning algorithms to obtain an optimal feature subset through the optimization of objective functions [10]. Compared to wrapper and embedded methods, filter methods are both handy and efficient. Therefore, this paper will focus on employing the filter method. Moreover, depending on different processing of multi-label data, multi-label feature selection methods can be classified into indirect methods and direct methods. Indirect methods are to convert multi-label data into single-label data, and subsequently applying single-label feature selection methods on the transformed data. However, indirect methods invariably neglect the correlation between labels. Direct methods directly select features using multi-label data, effectively considering the impact of the label space.

Information theory is a commonly employed evaluation method in direct methods, specifically in multi-label feature selection methods where correlation is taken into consideration. Scholars have proposed

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Fig. 1. The positive correlation between different candidate features and selected feature f_s under the label set L .

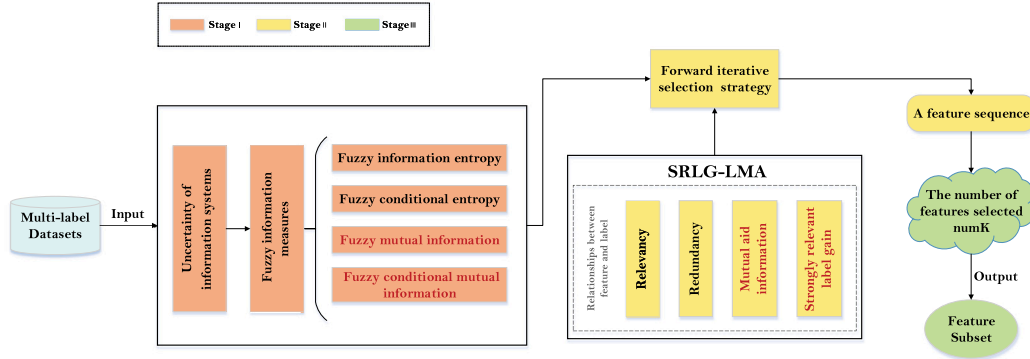


Fig. 2. The framework of this paper.

various effective methods for multi-label feature selection based on the entropy measures, and these methods have been verified to be feasible. Lee and Kim [11] designed a feature selection method based on multi-variate mutual information and approximate high-dimensional entropy for multi-label classification. Zhang et al. [12] proposed a global optimization framework that integrates feature correlation, redundancy and label correlation for considering multi-label feature selection. Lee and Kim [13] introduced a multi-label feature selection method based on the mutual information between features and labels and optimized the complexity of the algorithm. Li et al. [14] presented a multi-label granularity feature selection method based on the maximal correlation and minimal redundancy of mutual information. Lim et al. [15] defined a feature selection scoring function based on mutual information and proposed a feature selection method based on numerical optimization. Dai et al. [7] designed two new multi-label feature selection methods based on label symmetry uncertainty and feature redundancy assessment. In fact, most of the aforementioned studies primarily focus on analyzing the correlation between features and labels, as well as identifying redundancy between features. However, in multi-label data, the number of labels is multiple, and the internal influence of label space cannot be ignored. Actually, fuzzy conditional mutual information based on fuzzy rough sets can process numerical data and fully consider the mutual aid information between labels. Therefore, we take a comprehensive look at the influences between features and labels, the influences between features, and the influences between labels. Additionally, we assess if there is a positive correlation between features besides redundancy. Some scholars used conditional mutual information to describe the interaction or complementarity information between features [16,17]. Although these studies all deeply consider the positive correlation between features, there may still be a problem in the process of multi-label feature selection, and we use an example to illustrate this problem.

In Fig. 1, let $L = \{l_1, l_2\}$ be a set of labels, $K = \{f_1, f_2\}$ be a set of candidate features and $S = \{f_s\}$ be a set of selected features. We assume that $I(f_1; l_1 | f_s) = 0.75$, $I(f_1; l_2 | f_s) = 0.375$, $I(f_2; l_1 | f_s) = 0.5625$ and $I(f_2; l_2 | f_s) = 0.5625$. In the existing study of positive correlation between features, for candidate feature f_w , the larger the value of

$\sum_{i=1} I(f_w; l_i | f_s)$, it indicates that the candidate feature f_k relative to selected feature f_s provides additional information to the labels. Obviously, in the above example, $\sum_{i=1} I(f_1; l_i | f_s) = \sum_{i=1} I(f_2; l_i | f_s)$ indicates that we cannot select better features from the interaction between features. In the interaction between features, the candidate feature f_w will be considered as redundant relative to selected features when $I(f_w; l_i | f_s) \leq I(f_s; l_i)$ [16]. Then, from the point of view of multi-label data, it can be considered that the candidate feature f_w is weakly relevant with labels relative to selected features, and vice versa. At the same time, we can clearly see that the candidate feature f_w may have different relationships with labels relative to different selected features. Therefore, to further explore the positive correlation between features, we propose a novel multi-label feature selection method based on strongly relevant label gain and label mutual aid based on fuzzy information measures. The framework of this paper is presented in Fig. 2. The main contributions of our work can be summarized as follows:

- (1) This paper introduces two new relationships between labels and candidate features relative to selected features. Additionally, a label discriminant function is defined. Furthermore, the changes in the relationships of labels and candidate features relative to different selected features are investigated.
- (2) The mutual aid information between labels based on fuzzy conditional mutual information is presented, which can describe the potential correlation between labels in label space.
- (3) The strongly relevant label gain is proposed by the label discriminant function, which can capture the subset of labels with the highest degree of relevance to the candidate feature.
- (4) A novel multi-label feature selection algorithm via strongly relevant label gain and label mutual aid (called SRLG-LMA) is proposed. The effectiveness of the proposed method is demonstrated by comparing it with other representative multi-label feature selection algorithms on sixteen multi-label benchmark datasets.

The structure of the paper is as follows. Section 2 gives a brief review for the related work. Section 3 reviews several basic concepts of multi-label data, fuzzy similarity relations and fuzzy information measures. Section 4 proposes two new relationships between features

and labels relative to selected features via fuzzy information measures, and analyze the change of relationship between features and labels relative to different selected features. Section 5 introduces label mutual aid and strongly relevant label gain, and proposes a novel multi-label feature selection method. Section 6 compares the proposed algorithm with other representative multi-label feature selection algorithms on the multi-label benchmark datasets. Finally, this work is summarized in Section 7.

2. Related work

Multi-label data, which is widely used in the real world, has been intensively studied in the field of feature selection. There are two categories of existing multi-label feature selection methods: indirect methods and direct methods. Indirect methods are to transform multi-label data into single-label data, and then use single-label feature selection to handle the transformed data which is also called problem transformation method, such as the Pruned Problem Transformation (PPT) [18], Binary Relevance (BR) [19], and Label Power set (LP) [20]. Read et al. [18] used χ^2 statistic to select important features together with PPT (PPT+CHI). Bouttelle et al. [19] proposed a pattern recognition framework for processing multi-label data, and introduced a new metric for evaluating various indicators of a single example. Tsoumakas et al. [20] proposed a novel multi-label learning method based on label powerset (LP), which randomly granulates the initial label set into multiple label subsets, and then uses the LP method to train the classifier. However, indirect methods consistently fail to consider label correlation, which is essential for accurate feature selection. Consequently, scholars have proposed various multi-label feature selection methods to tackle this issue by directly selecting relevant features based on multi-label data.

In direct methods, most scholars consider not only the correlation between labels, but also the correlation of features and labels in multi-label data. Huang et al. [21] designed a unified framework for feature selection and classification, which learns features for specific labels and shared features to distinguish each class label by exploiting the correlation between two labels, and builds a classifier on the multi-label data after feature selection. In order to capture label correlation locally and consider the application of label information to constrain the symbiotic relationship of label pairs, Zhang et al. [22] proposed a manifold regularization-based embedded multi-label feature selection algorithm, which constructs a low-dimensional embedding for matching the label distribution. Lim et al. [23] introduced a parameter-free population initialization method based on the multi-label feature selection of evolutionary algorithm (EA) considering the correlations between features and labels. Zhang et al. [24] presented a novel multi-label feature selection method based on group-preserving and label-specific, which performs feature selection considering both the special features of labels in the same set and the specific features that each label owns and considers the correlations of label group and instance group to improve model generalization. With the wide range of applications of information theory, the research on information theory based multi-label feature selection has developed rapidly. Lee and Kim [11] defined a feature selection method that applies multivariate mutual information to measure the correlation between feature and label and presented the approximate calculation method of high-dimensional joint entropy. Li et al. [14] proposed a granular feature selection strategy for multi-label data based on the criterion of max-correlation and min-redundancy which ensures that the class discrimination information of the selected feature subset is maximized and the internal redundancy is minimized. A scalable multi-label feature selection method based on mutual information correlation evaluation was proposed by Lee and Kim [25], which theoretically demonstrated the characteristics of correlation measurement method, and applied an approximation for measuring multi-label feature dependency. Lin et al. [26] introduced fuzzy mutual information for evaluating the

Table 1

A multi-label decision table framework.

U	f_1	f_2	f_3	$\dots$	f_d	l_1	$\dots$	l_m
u_1	x_{11}	x_{12}	x_{13}	$\dots$	x_{1d}	y_{11}	$\dots$	y_{1m}
u_2	x_{21}	x_{22}	x_{23}	$\dots$	x_{2d}	y_{21}	$\dots$	y_{2m}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
u_i	x_{i1}	x_{i2}	x_{i3}	$\dots$	x_{id}	y_{i1}	$\dots$	y_{im}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
u_n	x_{n1}	x_{n2}	x_{n3}	$\dots$	x_{nd}	y_{n1}	$\dots$	y_{nm}

significance of features in multi-label feature selection, and designed efficient non-streaming feature selection algorithm and streaming feature selection algorithm. Dai et al. [7] proposed the first multi-label feature selection approach via the symmetric uncertainty between features and labels. Considering the redundancy between features, the second multi-label feature selection approach is proposed by introducing feature redundancy processing. Zhang et al. [27] defined three types of label relationships based on the mutual information and conditional mutual information, and then proposed two novel feature selection methods called LSMFS and MLSMFS by analyzing the change of label relationships.

In fact, most existing studies overlook the mutual aid information between labels. Meanwhile, in the multi-label data, the existing multi-label feature selection methods seldom consider the positive correlation between features, which may lead to poor classification performance of the selected features. Therefore, this paper presents a novel multi-label feature selection method that addresses the two aforementioned issues, by utilizing label mutual aid and strongly relevant label gain.

3. Preliminary knowledge

In this section, we will briefly review some basic notations of multi-label data, fuzzy similarity relations and uncertainty measures based on entropy and mutual information.

3.1. Multi-label data

Suppose that $MLDT = \langle U, F \cup L, V \rangle$ is a Multi-Label Decision Table. $U = \{u_1, u_2, \dots, u_n\}$ represents a nonempty finite set of samples; $F = \{f_1, f_2, \dots, f_d\}$ is a nonempty finite set of features; $L = \{l_1, l_2, \dots, l_m\}$ is a nonempty finite set of labels; $V = \{(u_i, x_i, y_i) | u_i \in U\}$ is the domain and range of multi-label data. For $\forall u_i \in U$, the d -dimensional feature vector x_i follows that $x_i = \{x_{i1}, x_{i2}, \dots, x_{id}\}$ where x_{id} is a feature value, and the m -dimensional label vector y_i follows that $y_i = \{y_{i1}, y_{i2}, \dots, y_{im}\}$ where $y_{im} \in \{0, 1\}$. If $y_{im} = 1$, then the label l_m is associated with the sample u_i ; otherwise, the label l_m is not associated with the sample u_i . To better understand the above concepts, we present a multi-label decision table framework in Table 1.

3.2. Fuzzy similarity relations

Fuzzy similarity relations are utilized to describe the similarity between samples. Suppose that U is a non-empty universe, R is called a fuzzy similarity relation, for $\forall x, y, z \in U$, if R satisfies:

- (1) Reflexivity: $R(x, x) = 1$;
- (2) Symmetry: $R(x, y) = R(y, x)$;
- (3) Transitivity: $R(x, z) \geq T\{R(x, y), R(y, z)\}$.

The T is a T -norm, it is defined as follows:

Definition 3.1 ([28]). Suppose that $P = [0, 1]$, for $\forall u, v, w, z \in P$, the mapping $T : P \times P \rightarrow P$ satisfies the coming properties:

- (1) Boundary condition: $T(1, u) = u$;
- (2) Commutativity: $T(u, v) = T(v, u)$;
- (3) Associativity: $T(u, T(v, w)) = T(T(u, v), w)$;
- (4) Monotonicity: $w \geq u, z \geq v \Rightarrow T(u, v) \leq T(w, z)$.

In practical applications, the use of classical fuzzy operators for handling fuzzy data often results in a significant loss of information, which makes the treatment of the problem out of reality. The problem can be solved by using T -norms. The frequently used T -norms are as follows:

- (1) $T_M(u, v) = \min\{u, v\}$;
- (2) $T_{\cos}(u, v) = \max\{uv - \sqrt{1-u^2}\sqrt{1-v^2}, 0\}$.

Theorem 3.2 ([28]). For $\forall u, v \in [0, 1]$, $T_M(u, v) \geq T_{\cos}(u, v)$.

From Theorem 3.2, it is clear that the fuzzy relation structured by T_M is more stricter. Apparently, if $T = T_M$, then R is a fuzzy equivalence relation. However, the construction of fuzzy equivalence relations is more challenging in practical applications. Therefore, we normally use fuzzy $T_{\cos}$ -similarity relation to represent the relationship between samples.

Given a nonempty universe U , $\forall u, v \in U$ and $P, Q \subseteq F$, some operations on fuzzy $T_{\cos}$ -similarity relation R are defined as follows:

- (1) $R_P = R_Q \Leftrightarrow R_P(u, v) = R_Q(u, v)$;
- (2) $R = R_P \cap R_Q \Leftrightarrow R(u, v) = R_P(u, v) \wedge R_Q(u, v)$;
- (3) $R = R_P \cup R_Q \Leftrightarrow R(u, v) = R_P(u, v) \vee R_Q(u, v)$;
- (4) $R_P \subseteq R_Q \Leftrightarrow R_P(u, v) \leq R_Q(u, v)$.

Given a non-empty universe U , R is a fuzzy $T_{\cos}$ -similarity relation over U , denoted by a relation matrix $M(R)$:

$$M(R) = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{nn} \end{pmatrix}, \quad (1)$$

where r_{ij} can be represented as $R(x_i, x_j) = \exp\left(\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$ and $R(x_i, x_j) \in [0, 1]$.

$R(x_i, x_j)$ can be used to measure the degree of similarity between two samples x_i and x_j . If x_i is totally equal to x_j , then $R(x_i, x_j) = 1$, while x_i is entirely different from x_j , then $R(x_i, x_j) = 0$.

Definition 3.3 ([29]). Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, $B \subseteq F$ is a feature subset. With the fuzzy $T_{\cos}$ -similarity relation R_B induced by the feature subset B , the nonempty set U can be divided into granularities as follows:

$$U/R_B = \{[x_i]_{R_B}\}_{i=1}^n, \quad (2)$$

where $[x_i]_{R_B} = \frac{R_B(x_i, x_1)}{x_1} + \frac{R_B(x_i, x_2)}{x_2} + \cdots + \frac{R_B(x_i, x_n)}{x_n}$ is the fuzzy $T_{\cos}$ -similarity class generated by the sample x_i , and $R_B(x_i, x_j)$ describes the similarity between sample x_i and sample x_j in the feature space formed by B .

The cardinality of $[x_i]_{R_B}$ is designed as $|[x_i]_{R_B}| = \sum_{j=1}^n R_B(x_i, x_j)$, which reflects the ability of fuzzy $T_{\cos}$ -similarity relation to divide the universe. Since R is the fuzzy $T_{\cos}$ -similarity relation, then $[x_i]_{R_B}$ is a fuzzy set.

3.3. Fuzzy information measures

Fuzzy entropy and fuzzy mutual information are commonly employed to measure uncertainty in information systems. Some relevant definitions of fuzzy information theory are presented below:

Definition 3.4 ([29]). Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, for the feature subset $B \subseteq F$, the fuzzy similarity relation on B is expressed as R_B , then the fuzzy information entropy of B is designed as follows:

$$MLH(R_B) = -\frac{1}{n} \sum_{i=1}^n \log_2 \frac{|[x_i]_{R_B}|}{n}, \quad (3)$$

where $[x_i]_{R_B}$ is the fuzzy similarity class of x_i induced by R_B and the $|[x_i]_{R_B}|$ is the cardinality of fuzzy similarity class $[x_i]_{R_B}$.

According to Definition 3.4, the greater the distinguishing ability of a feature set, the larger the value of the corresponding fuzzy information entropy, which is significant for feature selection.

Definition 3.5 ([29]). Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, for the feature subsets $B, D \subseteq F$, the fuzzy conditional entropy of B about D is designed as follows:

$$MLH(R_B|R_D) = -\frac{1}{n} \sum_{i=1}^n \log_2 \frac{|[x_i]_{R_B} \cap [x_i]_{R_D}|}{|[x_i]_{R_D}|}. \quad (4)$$

Fuzzy conditional entropy is employed to measure the uncertainty between features and labels. It is commonly utilized in the process of feature selection [30].

To better describe the uncertainty between features and between features and labels, the concepts of fuzzy mutual information and fuzzy conditional mutual information are introduced based on fuzzy $T_{\cos}$ -similarity relation R . The specific definitions are as follows:

Definition 3.6 ([31]). Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, for the feature subsets $B, D \subseteq F$, the fuzzy mutual information between feature subsets B and D is designed as follows:

$$MLI(R_B; R_D) = \frac{1}{n} \sum_{i=1}^n \log_2 \frac{n \cdot |[x_i]_{R_B} \cap [x_i]_{R_D}|}{|[x_i]_{R_D}| \cdot |[x_i]_{R_B}|}. \quad (5)$$

Definition 3.7 ([31]). Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, for the feature subsets $B, D, E \subseteq F$, the fuzzy conditional mutual information between feature subsets B and D about E is designed as follows:

$$MCLI(R_B; R_D|R_E) = \frac{1}{n} \sum_{i=1}^n \log_2 \frac{|[x_i]_{R_B} \cap [x_i]_{R_D} \cap [x_i]_{R_E}| \cdot |[x_i]_{R_E}|}{|[x_i]_{R_B} \cap [x_i]_{R_E}| \cdot |[x_i]_{R_D} \cap [x_i]_{R_E}|}. \quad (6)$$

4. New relationships between features and labels via fuzzy information measures

In this section, Section 4.1 introduces two new relationships between candidate features and labels relative to selected features. Additionally, Section 4.2 analyzes the change of relationship between candidate features and labels relative to different selected features, and presents a novel discriminant function that prioritizes the subset of labels most relevant to the candidate features.

4.1. Two new relationships between a candidate feature and a label relative to a selected feature

In general, the relevance between labels and features is considered in multi-label feature selection [2,7]. However, the relationships between a candidate feature and a label relative to a selected feature are frequently disregarded. It is apparent from the analysis of Fig. 1 that the relationship between a label and different candidate features relative to selected features is not exactly the same. Based on the above analysis, we define two new relationships between a candidate feature and a label relative to a selected feature.

By calculating the difference of different candidate features for the same label relative to a selected feature, labels can be categorized into two types: strongly relevant label and weakly relevant label.

Definition 4.1 (Strongly Relevant Label). Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, $f_s \in S$ is a selected feature where $S \subseteq F$, $f_w \in F - S$ is a candidate feature. $l_i \in L$ is a label. If $MCLI(f_w; l_i|f_s) - MLI(f_s; l_i) > 0$, then label l_i is strongly relevant to candidate feature f_w relative to selected feature f_s .

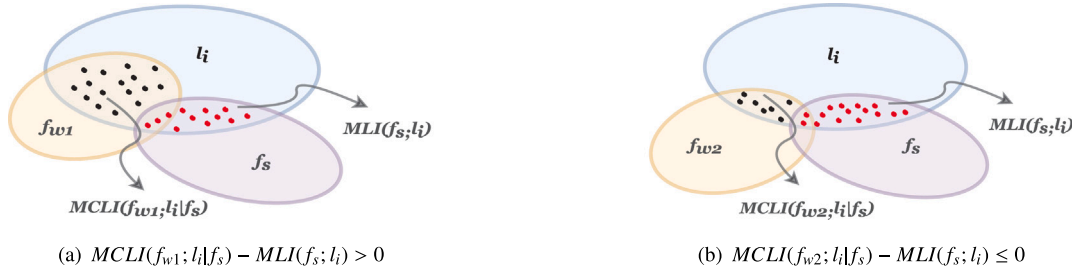


Fig. 3. The relationships between a candidate feature and a label relative to a selected feature.

Fig. 3(a) illustrates that the part $MCLI(f_{w1}; l_i | f_s)$ is greater than $MLI(f_s; l_i)$, indicating $MCLI(f_{w1}; l_i | f_s) - MLI(f_s; l_i) > 0$. This implies that the amount of shared classification information between label l_i and candidate feature f_{w1} relative to selected feature f_s is greater compared to that between selected feature f_s and label l_i . In other words, the label l_i is strongly relevant to the candidate feature f_{w1} when given the selected feature f_s .

Definition 4.2 (Weakly Relevant Label). Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, $f_s \in S$ is a selected feature where $S \subseteq F$, $f_w \in F - S$ is a candidate feature. $l_i \in L$ is a label. If $MCLI(f_w; l_i | f_s) - MLI(f_s; l_i) \leq 0$, then label l_i is weakly relevant to candidate feature f_w relative to selected feature f_s .

In Fig. 3(b), it can be observed that $MCLI(f_{w2}; l_i | f_s) - MLI(f_s; l_i) \leq 0$. This inequality indicates that the classification information between label l_i and candidate feature f_{w2} relative to selected feature f_s is smaller than the classification information between selected feature f_s and label l_i . Thus, we infer a weakly relevant relationship between label l_i and candidate feature f_{w2} .

According to Definitions 4.1 and 4.2, it is evident that the relationships between a candidate feature and a label are relative, not absolute. For example, in Fig. 3, the same label has different degrees of relevance for different candidate features relative to a selected feature. Similarly, different labels have different degrees of relevance for the same candidate feature relative to a selected feature. Therefore, we define a discriminant function to measure the degrees of relevance between a label and a candidate feature, which is defined as follows:

Definition 4.3. Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, $f_s \in S$ is a selected feature where $S \subseteq F$, $f_w \in F - S$ is a candidate feature. $l_i \in L$ is a label. Then the Discriminant Function between l_i for f_w relative to f_s is defined as:

$$DF(f_w, l_i, f_s) = MCLI(f_w; l_i | f_s) - MLI(f_s; l_i). \quad (7)$$

From Definition 4.3, if label l_i is strongly relevant to the candidate feature f_w relative to the selected feature f_s , then $DF(f_w, l_i, f_s)$ has a value greater than 0. Conversely, if label l_i is weakly relevant to the candidate feature f_w relative to the selected feature f_s , then $DF(f_w, l_i, f_s)$ is less than or equal to 0.

4.2. A new discriminant function between a label and a candidate feature relative to selected feature set

Remark 4.4. For $l_i \in L$, $\exists f_m, f_n \in S$ where $f_m \neq f_n$, when the selected feature f_s changes, the relationships between label l_i and candidate feature f_w relative to selected feature f_s will change.

For a label, different selected features can have different influence on the relationships between the label and a candidate feature. Specifically, when considering selected feature f_m , $DF(f_w, l_i, f_m) < 0$ may be obtained, while $DF(f_w, l_i, f_n) > 0$ may be obtained when considering selected feature f_n . To further affirm the accuracy of this conclusion, we employ the subsequent example to validate it.

Table 2

A multi-label decision table.

	f_1	f_2	f_3	f_4	l_1	l_2
x_1	0.36	0.06	0.39	0.11	1	1
x_2	0.40	0.25	0.84	0.66	0	1
x_3	0.07	0.25	0.70	0.42	1	0

Table 3

Supplemental gain under different labels.

Under label l_1	Under label l_2
$DF(f_1, l_1, f_2) = 0.1126$	$DF(f_1, l_2, f_2) = 0.1751$
$DF(f_1, l_1, f_3) = -0.1416$	$DF(f_1, l_2, f_3) = 0.2472$
$DF(f_1, l_1, f_4) = -0.3199$	$DF(f_1, l_2, f_4) = 0.2456$

Example 4.5. Give a multi-label decision table by Table 2, where $U = \{x_1, x_2, x_3\}$, $B = \{f_1, f_2, f_3, f_4\}$, $L = \{l_1, l_2\}$. Take $R_f(x_i, x_j) = \exp\left(\frac{\|f(x_i) - f(x_j)\|^2}{2\sigma^2}\right)$, where $f(x_i)$ represents the value of sample x_i on feature f , and $R_B = \cap_{f \in B} R_f$. Then, the fuzzy relation matrix of features and labels can be computed as:

$$\begin{aligned}
 M(R_{f_1}) &= \begin{pmatrix} 1.0000 & 0.8146 & 0.1462 \\ 0.8146 & 1.0000 & 0.4181 \\ 0.1462 & 0.4181 & 1.0000 \end{pmatrix} \\
 M(R_{f_2}) &= \begin{pmatrix} 1.0000 & 0.2231 & 0.2231 \\ 0.2231 & 1.0000 & 1.0000 \\ 0.2231 & 1.0000 & 1.0000 \end{pmatrix} \\
 M(R_{f_3}) &= \begin{pmatrix} 1.0000 & 0.1482 & 0.4041 \\ 0.1482 & 1.0000 & 0.8313 \\ 0.4041 & 0.8313 & 1.0000 \end{pmatrix} \\
 M(R_{f_4}) &= \begin{pmatrix} 1.0000 & 0.1368 & 0.3315 \\ 0.1368 & 1.0000 & 0.6847 \\ 0.3315 & 0.6847 & 1.0000 \end{pmatrix} \\
 M(R_{l_1}) &= \begin{pmatrix} 1.0000 & 0.0000 & 1.0000 \\ 0.0000 & 1.0000 & 0.0000 \\ 1.0000 & 0.0000 & 1.0000 \end{pmatrix} \\
 M(R_{l_2}) &= \begin{pmatrix} 1.0000 & 1.0000 & 0.0000 \\ 1.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.0000 & 1.0000 \end{pmatrix}
 \end{aligned}$$

By combining the fuzzy similarity matrix, we can calculate the corresponding DF for the candidate features under different labels in Table 3. As shown in Table 3, for label l_2 , it is evident that $DF > 0$ holds true for different selected features, which is the expected result. However, it is clear that there are variations in the relationships between label l_i and candidate feature f_w depending on the selected feature f_s . For example, under the selected feature f_2 , the information shared between candidate feature f_1 and label l_1 is $DF(f_1, l_1, f_2) = 0.1126$, indicating that $DF > 0$. But under the selected features f_3 , the information shared between f_1 and l_1 is $DF(f_1, l_1, f_3) = -0.1416$, indicating that $DF < 0$. Thus, label l_1 is strongly relevant to candidate feature f_1 relative to selected feature f_2 , but label l_1 is weakly relevant to candidate feature f_1 relative to selected feature f_3 .

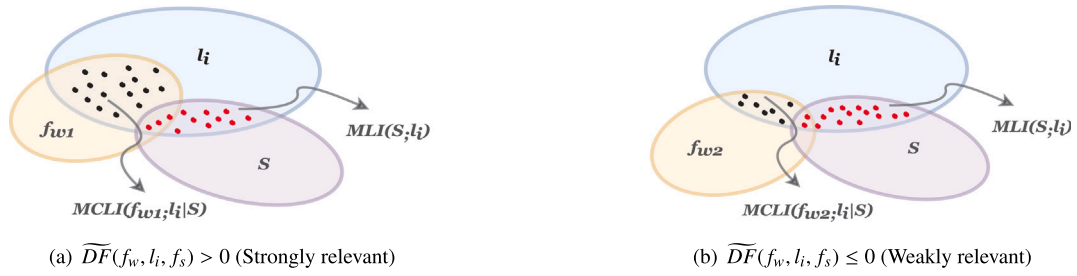


Fig. 4. The relationship between a candidate feature and a label relative to selected feature set.

The relationships between a label and a candidate feature can be strongly or weakly relevant depending on the selected feature. Therefore, the impact of selected feature variations should not be overlooked during the computational process. Based on the above consideration, we propose a different approach to calculate the discriminant function. More specifically, we redefine the calculation method from a new perspective. This new discriminant function is defined as follows:

Definition 4.6. Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, $U = \{x_1, x_2, \dots, x_n\}$ represents a set with n samples, $F = \{f_1, f_2, \dots, f_d\}$ denotes a set of features, $L = \{l_1, l_2, \dots, l_m\}$ is a set of labels, S is the a of selected features, $f_w \in F - S$ is a candidate feature. Based on the selected feature set S , the Discriminant Function between l_i and f_w is defined as:

$$\begin{aligned} \widetilde{DF}(f_w, l_i, S) &= MCLI(f_w, l_i | S) - \frac{MLI(S; l_i)}{|S|} \\ &= \frac{1}{n} \sum_{i=1}^n \log_2 \frac{|[x_i]_{f_w} \cap [x_i]_{l_i} \cap [x_i]_S| \cdot |[x_i]_S|^2 \cdot |[x_i]_{l_i}|}{n \cdot |[x_i]_{f_w} \cap [x_i]_S| \cdot |[x_i]_{l_i} \cap [x_i]_S|^2}. \end{aligned} \quad (8)$$

Theorem 4.7. For $\forall l_i, l_j \in L$, if $\widetilde{DF}(f_w, l_i, S) \geq \widetilde{DF}(f_w, l_j, S)$, the label l_i is more relevant than the label l_j and candidate feature f_w relative to selected feature set S . More precisely, if $\widetilde{DF}(f_w, l_i, S) > \widetilde{DF}(f_w, l_j, S)$, the label l_i is more strictly relevant than the label l_j and candidate feature f_w .

Proof. From Definitions 4.1 and 4.2, if label l_i and label l_j are strongly relevant labels and $\widetilde{DF}(f_w, l_i, S) \geq \widetilde{DF}(f_w, l_j, S)$ is satisfied, then it can be proved that label l_i is more relevant than label l_j with respect to candidate feature f_w . Similarly, when l_i and l_j are both weakly relevant labels, the above conclusion still holds. If l_i is a strongly relevant label and l_j is a weakly relevant label, $\widetilde{DF}(f_w, l_i, S) > \widetilde{DF}(f_w, l_j, S)$ always holds. In this case, l_i is considered to be more strictly relevant than label l_j with respect to candidate feature f_w .

In Formula (8), the first item represents the amount of effective information for classification between candidate feature f_w and label l_i relative to selected feature set S . The second item $MLI(S; l_i)$ measures the effective classification information between selected feature set S and label l_i . To ensure comparability, we compute the average value of $MLI(S; l_i)$ under S . The difference between these two items provides the gain of the strongly relevant label for candidate feature f_w . Since the second term changes dynamically as the label changes, we achieve a more equitable assessment of the relationships between the label and the candidate feature relative to the selected feature set.

Definition 4.6 is used to discriminate the relationships between candidate feature f_w and label l_i relative to selected feature set S . Fig. 4 displays the relationships between label l_i and candidate feature f_w . Clearly, $\widetilde{DF}(f_w, l_i, S) > 0$ means that label l_i is strongly relevant to candidate feature f_w , then label l_i is more important to candidate feature f_w than other labels. In other words, candidate feature f_w will provide more classification information for label l_i . $\widetilde{DF}(f_w, l_i, S) < 0$ means that label l_i is weakly relevant to candidate feature f_w , then it is considered to candidate feature f_w provide less classification information for label l_i than strongly relevant labels.

5. Strongly relevant label gain and label mutual aid via fuzzy information measures

In this section, based on the criterions of the maximum relevance and minimum redundancy [2], we define them from the perspective of fuzzy mutual information, and propose label mutual aid to explore the potential correlation in the label space. Furthermore, based on the work in Section 4, we introduce strongly relevant label gain to exploit the positive correlation between features, and propose a multi-label feature selection algorithm based on strong relevant label gain and label mutual aid.

5.1. Relevance between features and labels

In multi-label feature selection, a single feature may contain the classification information for multiple labels. Fuzzy mutual information can measure the amount of information shared between features and labels, so we define the relevance between candidate features and labels with fuzzy mutual information.

Definition 5.1. Given a multi-label decision table $MLDT = \langle U, F \cup L, V \rangle$, $U = \{x_1, x_2, \dots, x_n\}$ represents a set with n samples, $F = \{f_1, f_2, \dots, f_d\}$ denotes a set of features, $L = \{l_1, l_2, \dots, l_m\}$ is a label set, $S \subseteq F$ is a selected feature subset, and $f_w \in F - S$ is a candidate feature. Then, the relevance between candidate feature f_w and label l_i is defined as follows:

$$MREL(f_w, l_i) = MLI(f_w, l_i) = \frac{1}{n} \sum_{i=1}^n \log_2 \frac{n \cdot |[x_i]_{f_w} \cap [x_i]_{l_i}|}{|[x_i]_{l_i}| \cdot |[x_i]_{f_w}|}. \quad (9)$$

Theorem 5.2. If f_w and l_i are independent, then the relevance $MREL(f_w, l_i) = 0$. If l_i is completely dependent on f_w , then $MREL(f_w, l_i) = MLH(R_B)$.

Proof. By Definitions 3.4 and 3.5, we can get $MREL(f_w, l_i) = MLH(l_i) - MLH(l_i|f_w)$. Therefore, when candidate feature f_w and label l_i are independent of each other, $MLH(l_i|f_w) = MLH(l_i)$ can be obtained, then $MREL(f_w, l_i) = 0$. If l_i is completely dependent on f_w , we have $MLH(l_i|f_w) = 0$, then $MREL(f_w, l_i) = MLH(l_i)$. It is apparent that there is a maximum relevance between f_w and l_i .

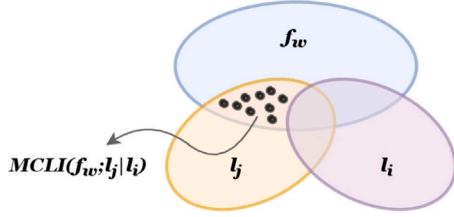
Based on Theorem 5.2, it is evident that $MREL(f_w, l_i)$ can serve as a measure of relevance between candidate features and labels. If candidate features and labels produce a greater amount of valid classification information, the value of $MREL(f_w, l_i)$ will be higher, and vice versa. Hence, in the process of feature selection, our preference lies in selecting the candidate feature f_w that yields the highest $\sum_{l_i \in L} MREL(f_w, l_i)$ value.

Example 5.3. Continue to Table 2, by utilizing the fuzzy similarity matrix, one can calculate the mutual information between candidate feature f_w and label l_i , with the resulting values presented in Table 4.

Table 4

The mutual information between candidate features and labels.

	f_1	f_2	f_3	f_4
l_1	0.1243	0.1662	0.3180	0.4111
l_2	0.5662	0.1662	0.1245	0.1244

**Fig. 5.** Mutual aid information between labels.

Therefore, for feature f_1 , the mutual information with whole label set is $\sum_{l_i \in L} MREL(f_1, l_i) = 0.6905$. Similarly, we can obtain $\sum_{l_i \in L} MREL(f_2, l_i) = 0.3324$, $\sum_{l_i \in L} MREL(f_3, l_i) = 0.4425$, $\sum_{l_i \in L} MREL(f_4, l_i) = 0.5355$. Based on the results obtained from the mutual information analysis, it is evident that the candidate feature f_1 has the highest mutual information with the entire label set. Therefore, the selected feature set is $\{f_1\}$ and the candidate feature set is $\{f_2, f_3, f_4\}$.

5.2. Redundancy between features

However, it is not reasonable to solely focus on the relevance between candidate features and labels. There may be redundant overlapping classification information between candidate features and selected features. Consequently, we measure the amount of redundant information between candidate features and selected features using fuzzy mutual information, as defined in the following formula.

Definition 5.4. Given a multi-label decision table $MLDT = (U, F \cup L, V)$, $U = \{x_1, x_2, \dots, x_n\}$ represents a set with n samples, $F = \{f_1, f_2, \dots, f_d\}$ denotes a set of features, S is a selected feature set, $f_w \in F - S$ is a candidate feature. Then, the redundancy between f_w and S is defined as follows:

$$MRED(f_w, S) = MLI(f_w; S) = \frac{1}{n} \sum_{i=1}^n \log_2 \frac{n \cdot |[x_i]_{f_w} \cap [x_i]_S|}{|[x_i]_S| \cdot |[x_i]_{f_w}|}. \quad (10)$$

Similar to Theorem 5.2, the redundant information reflects the amount of overlapping classification information between candidate feature f_w and selected feature set S . In this case, lower $MRED(f_w, S)$ values indicate less overlapping information. Hence, the candidate feature f_w with the smallest $MRED(f_w, S)$ value should be selected.

Example 5.5. Continue to Example 5.3, suppose that the selected feature set is $\{f_1\}$, and the candidate feature set is $\{f_2, f_3, f_4\}$. By utilizing the fuzzy similarity matrix, we can obtain the fuzzy mutual information matrix between features, as shown below.

$$M(F \times F) = \begin{pmatrix} 0.6597 & 0.3184 & 0.2881 & 0.2898 \\ 0.3184 & 0.6391 & 0.5389 & 0.4688 \\ 0.2881 & 0.5389 & 0.6583 & 0.5916 \\ 0.2898 & 0.4688 & 0.5916 & 0.6677 \end{pmatrix}.$$

Based on the provided matrix, we can determine the redundant information between candidate feature f_2 and selected feature set $S = \{f_1\}$, which is $MRED(f_2, S) = 0.3184$. Similarly, we can obtain that $MRED(f_3, S) = 0.2881$ and $MRED(f_4, S) = 0.2898$.

Table 5

The amount of mutual aid information.

	f_1	f_2	f_3	f_4
$l_2 l_1$	0.5354	0.4729	0.3402	0.2567
$l_1 l_2$	0.0935	0.4729	0.5337	0.5434

5.3. Mutual aid information between labels

In the preceding discussion, we examined the relevance between candidate features and labels, as well as the redundancy between candidate features and selected features. However, the impact of label mutual aid on reducing uncertain classification information has been disregarded. As a result, we introduce the concept of label mutual aid based on fuzzy information measure, which is defined as follows:

Definition 5.6. Given a multi-label decision table $MLDT = (U, F \cup L, V)$, $U = \{x_1, x_2, \dots, x_n\}$ represents a set with n samples, $F = \{f_1, f_2, \dots, f_d\}$ denotes a set of features, $L = \{l_1, l_2, \dots, l_m\}$ is a label set, $S \subseteq F$ is a selected feature set, $f_w \in F - S$ is a candidate feature, $l_j \in L - \{l_i\}$. The amount of mutual aid information of l_j for f_w under l_i is defined as:

$$MINT(f_w, l_i, l_j) = MCLI(f_w; l_j | l_i) = \frac{1}{n} \sum_{i=1}^n \log_2 \frac{|[x_i]_{f_w} \cap [x_i]_{l_j} \cap [x_i]_{l_i}| \cdot |[x_i]_{l_i}|}{|[x_i]_{f_w} \cap [x_i]_{l_i}| \cdot |[x_i]_{l_j} \cap [x_i]_{l_i}|}. \quad (11)$$

$MINT(f_w, l_i, l_j)$ is employed to calculate the mutual aid information between candidate feature f_w and label l_j under the given label l_i . The mutual aid information between label l_i and label l_j is illustrated in Fig. 5. It is evident that, under the given label l_i , the additional information of both candidate feature f_w and label l_j is a result of the mutual aid between label l_i and label l_j .

The label mutual aid allows for a deeper exploration of the potential relationships between labels. The amount of mutual aid information is different under different candidate features. Consequently, the larger the value of $MINT(f_w, l_i, l_j)$, the more effectively the candidate feature f_w can reduce classification uncertainty. We demonstrate the validity of the above conclusion using Example 5.7.

Example 5.7. Continue to Example 5.3, by utilizing the fuzzy similarity matrix of features and labels, the amount of mutual aid information can be calculated by Definition 5.6. The results are shown in Table 5.

Thus, the amount of mutual aid information for the candidate feature f_2 can be obtained from Table 5. It is calculated as $\sum_{l_i \in L} \sum_{l_j \in L - \{l_i\}} MINT(f_2, l_i, l_j) = 0.9458$. Similarly, we can obtain $\sum_{l_i \in L} \sum_{l_j \in L - \{l_i\}} MINT(f_3, l_i, l_j) = 0.8739$ and $\sum_{l_i \in L} \sum_{l_j \in L - \{l_i\}} MINT(f_4, l_i, l_j) = 0.8001$.

5.4. Strongly relevant label gain with respect to features

In the above study, although we have taken into account all the effects between candidate features and labels, between candidate features and selected features, between labels. Nevertheless, the positive correlation between features remains significant. Two significant relationships exist between labels and candidate features relative to selected feature set that are effective in reducing classification uncertainty.

Based on Definition 4.6, the relationships between label l_i and candidate feature f_w relative to selected feature set S are as follows:

$$l_i = \begin{cases} \text{Strongly relevant label,} & \widetilde{DF}(f_w, l_i, S) > 0, \\ \text{Weakly relevant label,} & \widetilde{DF}(f_w, l_i, S) \leq 0. \end{cases} \quad (12)$$

According to Formula (12), $\widetilde{DF}(f_w, l_i, S)$ is used to identify the relationships between label l_i and candidate feature f_w relative to selected feature set S . If $\widetilde{DF}(f_w, l_i, S) > 0$, then it indicates that the relationship

between l_i and f_w relative to selected feature set S is strongly relevant. If $\widetilde{DF}(f_w, l_i, S) \leq 0$, then the relationship between l_i and f_w relative to selected feature set S is weakly relevant. $\widetilde{DF}(f_w, l_i, S)$ shows the value of label gain information. Based on the above analysis, a correlation term considering strongly relevant label gain is proposed.

Definition 5.8. Given a multi-label decision table $MLDT = (U, F \cup L, V)$, $U = \{x_1, x_2, \dots, x_n\}$ represents a set with n samples, $F = \{f_1, f_2, \dots, f_d\}$ denotes a set of features, $L = \{l_1, l_2, \dots, l_m\}$ is a label set, $S \subseteq F$ is a selected feature set, $f_w \in F - S$ is a candidate feature. The strongly relevant label gain is defined as:

$$SRLG(f_w, l_i, S) = \max\{\widetilde{DF}(f_w, l_i, S), 0\}. \quad (13)$$

$SRLG(f_w, l_i, S)$ computes the gain information between candidate feature f_w and label l_i relative to selected feature set S . If $\widetilde{DF}(f_w, l_i, S) \leq 0$, it indicates that the relationship between label l_i and candidate feature f_w relative to selected feature set S is weakly relevant. Therefore, $SRLG(f_w, l_i, S)$ can effectively reinforce the influence of higher relevant labels to the candidate feature relative to the selected feature set and weaken the influence of weaker relevant labels to the candidate feature relative to the selected feature set.

According to Definition 5.8, the problem raised in the introduction can be effectively solved. As shown in Fig. 1, assuming $I(l_1; S) = I(l_2; S) = 0.6$, it is possible to conclude that $SRLG(f_1, l_1, S) + SRLG(f_1, l_2, S) > SRLG(f_2, l_1, S) + SRLG(f_2, l_2, S)$. Clearly, in the case of strongly relevant label gain, f_1 can be identified as the superior feature.

5.5. Multi-label feature selection method using strongly relevant label gain and label mutual aid

Minimum redundancy, maximum relevance, strongly relevant label gain and label mutual aid based on fuzzy technology have been defined in the previous sections. Based on the above discussions, we propose a multi-label feature selection method.

Given a multi-label decision table $MLDT = (U, F \cup L, V)$, $U = \{x_1, x_2, \dots, x_n\}$, $F = \{f_1, f_2, \dots, f_d\}$, $L = \{l_1, l_2, \dots, l_m\}$, $f_s \in S$ is a selected feature, $f_w \in F - S$ is a candidate feature, $l_j \in L - \{l_i\}$. The proposed multi-label feature selection method based on fuzzy information measures is established as follows:

$$\begin{aligned} J(f_w) = & \frac{1}{|L|} \sum_{l_i \in L} \{MREL(f_w, l_i) + \sum_{l_j \in L - \{l_i\}} MINT(f_w, l_i, l_j) \\ & + SRLG(f_w, l_i, S)\} \\ & - \frac{1}{|S|} \sum_{l_i \in L} MRED(f_w, S). \end{aligned} \quad (14)$$

The importance of candidate feature f_w can be assessed by $J(f_w)$, as demonstrated by Formula (14). It is evident that candidate feature f_w must satisfy the maximum $J(f_w)$, as shown in the following equation:

$$f_w^\tau = \arg \max_{f_w \in F - S} J(f_w). \quad (15)$$

Example 5.9. Continue to Example 5.3, the $J(f_w)$ under candidate feature f_2 can be obtained from Formula (15), which is $J(f_2) = 0.4810$. Similarly, $J(f_3) = 0.5553$, $J(f_4) = 0.5670$. Therefore, we can get a sorted feature set, that is $\{f_1, f_4, f_3, f_2\}$.

The proposed multi-label feature selection method based on strongly relevant label gain and label mutual aid (SRLG-LMA) is implemented in the form of Algorithm 1, which calculates the corresponding importance of the candidate features according to Formula (15) and finally arranges the subset of features in descending order.

In algorithm 1, assuming the $|U|$ is the number of samples, $|F|$ is the number of features, $|L|$ is the number of labels, $|Selected|$ is the number

Algorithm 1 SRLG-LMA method based on fuzzy information measures.

Input: $MLIDS = (U, F \cup L, D)$, samples set $U = \{x_1, x_2, \dots, x_n\}$, features set $F = \{f_1, f_2, \dots, f_d\}$, labels set $L = \{l_1, l_2, \dots, l_m\}$ and the number of features selected numK.

Output: A reduct subset: *Selected*.

```

1: Initialize Selected  $\leftarrow \emptyset$ ;
2: Initialize unSelected  $\leftarrow \text{zeros}(1, d)$ ;
3: for  $i = 1$  to  $d$  do
4:   Compute the relevance RelList( $i$ ) based on Definition 5.1;
5:   Compute the label mutual aid information LadList( $i$ ) based on Definition 5.6;
6: end for
7:  $f_1 \leftarrow \text{select max}\{\text{RelList}\}$ ;
8: for  $k = 1$  to  $\text{numK}$  do
9:    $\text{mit} \leftarrow \text{zeros}(1, d)$ 
10:  if  $k == 1$  then
11:    Selected  $\leftarrow \text{Selected} \cup \{f_1\}$ 
12:    unSelected  $\leftarrow \text{unSelected} - \{f_1\}$ 
13:  else
14:    for  $i = 1$  to  $d$  do
15:      if unSelected( $i$ ) == 0 then
16:        Compute the redundancy RedList( $i$ ) based on Definition 5.4;
17:        Compute the strongly relevant label gain SrlgList( $i$ ) based on Definition 5.8;
18:         $\text{mit}(i) \leftarrow \text{Obtain } J(f_i) \text{ by Formula (14)}$ 
19:      end if
20:    end for
21:     $f_j \leftarrow \text{select max}\{\text{mit}\}$ 
22:    Selected  $\leftarrow \text{Selected} \cup \{f_j\}$ 
23:    unSelected  $\leftarrow \text{unSelected} - \{f_j\}$ 
24:  end if
25: end for
26: return Selected;

```

of selected features, $|\text{numK}|$ is the threshold of selected features. All features in F are calculated in sequence and the feature with the best effect is selected. Step 1 to step 2 initialize the selected feature set and candidate feature set, which is the basis of the next step. Step 3 to step 6 calculate the relevance and the label mutual aid information between each feature and the label set, the time complexity is $O(|F||L|^2)$. Step 10 to step 12 are to select the first feature with the strongest relevance. Step 14 to step 18 are computing $J(f_w)$ of candidate features, the time complexity is $O(|\text{numK}||F||L|)$. Step 21 to step 23 are to select the candidate feature with the maximum value of $J(f_w)$ and store them in the selected feature set S . Step 26 returns the sorted feature set *Selected*.

6. Experiments and analyses

In this section, we validate the effectiveness of the proposed algorithm for multi-label feature selection and present the experimental results. Firstly, we present the definitions of multiple evaluation metrics for multi-label data in Section 6.1. Secondly, we provide the descriptions of multi-label benchmark datasets and experimental settings in Section 6.2. Finally, we demonstrate the effectiveness of the proposed algorithm SRLG-LMA in Section 6.3.

6.1. Evaluation metrics of multi-label feature selection

To more effectively evaluate the performance of multi-label algorithms, we have employed five commonly utilized evaluation metrics for multi-label learning [32,33]. Suppose $D = \{(x_i, y_i) | x_i \in X, y_i \in Y\}$ is a multi-label training data set, $U = \{x_1, x_2, \dots, x_n\}$ represents a set

Table 6
Descriptions of multi-label benchmark datasets.

Data sets	Instances	Training	Test	Features	Labels	Card	Dens	Dom
Emotions	593	391	202	72	6	1.869	0.311	Music
Birds	645	322	323	260	19	1.014	0.053	Audio
Yeast	2417	1500	917	103	14	4.237	0.303	Biology
Arts	5000	2000	3000	462	26	1.636	0.063	Text
Business	5000	2000	3000	438	30	1.588	0.053	Text
Computers	5000	2000	3000	681	33	1.508	0.046	Yahoo
Education	5000	2000	3000	550	33	1.461	0.044	Text
Health	5000	2000	3000	612	32	1.662	0.052	Text
Recreation	5000	2000	3000	606	22	1.423	0.065	Yahoo
Reference	5000	2000	3000	636	27	1.169	0.035	Yahoo
Society	5000	2000	3000	636	27	1.692	0.063	Text
Entertainment	5000	2000	3000	640	21	1.420	0.068	Text
Social	5000	2000	3000	1047	39	1.283	0.033	Text
Medical	978	333	645	1449	45	1.245	0.028	Text
Flags	194	129	65	19	7	3.392	0.485	Image
Scene	2407	1211	1196	294	6	1.074	0.179	Image

with n samples, $F = \{f_1, f_2, \dots, f_d\}$ denotes a set of features, $L = \{l_1, l_2, \dots, l_m\}$ is a label set. For $\forall x_i \in U$, $L(x_i)$, $L(x_i)'$ represent a true label set and a prediction label set, respectively. The specific definitions of evaluation metrics are as follows.

- **Hamming Loss (HL):** HL computes the frequency of occurrence which a given sample-label pair definitely is misclassified.

$$HL = \frac{1}{n} \sum_{i=1}^n \frac{|L(x_i) \Delta L(x_i)'|}{m}, \quad (16)$$

where Δ stands for the symmetric difference between true label set $L(x_i)$ and predicted label set $L(x_i)'$.

- **Average Precision (AP):** AP evaluates the average score of the label that ranks above a given label $l \in L$.

$$AP = \frac{1}{n} \sum_{i=1}^n \frac{1}{|L(x_i)|} \sum_{l \in L(x_i)} \frac{| \{l' | \text{rank}(x_i, l') \leq \text{rank}(x_i, l), l' \in L(x_i)\} |}{\text{rank}(x_i, l)}, \quad (17)$$

where $\text{rank}(\cdot, \cdot)$ stands for the ranking function, it denotes for two different labels $l_i, l_j \in L(x_i)$, if the evaluation function $f(x_i, l_i) > f(x_i, l_j)$, then $\text{rank}(x_i, l_i) < \text{rank}(x_i, l_j)$.

- **One Error (OE):** OE calculates the number of occurrences where the top-ranked label is not present in the appropriate multi-label set of a sample x_i .

$$OE = \frac{1}{n} \sum_{i=1}^n \ll [\arg \max_{l \in L(x_i)} f(x_i, l)] \notin L(x_i) \gg, \quad (18)$$

if any predicate κ holds, then $\ll \kappa \gg = 1$ and 0 otherwise.

- **Ranking Loss (RL):** RL evaluates the average score of label pairs in reverse sort for a given sample x_i .

$$RL = \frac{1}{n} \sum_{i=1}^n \frac{1}{|L(x_i)| \|\bar{L}(x_i)\|} | \{ (l', l'') | f(x_i, l') \leq f(x_i, l''), (l', l'') \in L(x_i) \times \bar{L}(x_i) \} |, \quad (19)$$

where $\bar{L}(x_i)$ denotes the complementary set of true label set $L(x_i)$.

- **Coverage (CV):** CV evaluates how many steps it takes on average to move down the list of ranked labels, covering all relevant labels for sample x_i .

$$CV = \frac{1}{n} \sum_{i=1}^n \max_{l \in L(x_i)} \text{rank}(x_i, l) - 1, \quad (20)$$

where the meaning of $\text{rank}(\cdot, \cdot)$ is consistent with the one in the AP .

The evaluation metrics employed in the experiment follow the convention of considering smaller values for HL , OE , RL and CV to indicate better classification performance. Conversely, higher values for AP indicate better classification performance.

6.2. Multi-label benchmark datasets and experimental settings

We utilize a total of sixteen real-world multi-label benchmark datasets from the Mulan Library, a renowned multi-label database, to evaluate the performance of our proposed multi-label feature selection algorithm. The specific details of these sixteen benchmark datasets are provided in Table 6. The table includes information on sample size, number of features, and label quantity, and encompassing various domains. The Medical dataset comprises 978 clinical free text reports labeled with one or more of 45 disease codes. The Flags dataset contains details of countries and their flags, aiming to predict some of the features. The Birds dataset can predict the set of bird species present, using a ten-second audio clip.

To highlight the superiority of the proposed algorithm, we conducted a comparative analysis with nine representative algorithms. The selected algorithms for comparison are as follows: PMU [11], MDDMproj [34], MDDMspc [34], MUCO [26], MCDM [35], SCLS [25], VMFS [36], GLFS [24], MDfS [22]. The parameters of each selected comparison algorithm are set as suggested by the corresponding literature.

The above algorithms can generate a sorted set of features. To determine the threshold for selected features, we set $NumK$ to 40% of the total number of features in the benchmark datasets. Furthermore, we employ the MLKNN classifier [37] to evaluate the performance based on five metrics: *Hamming Loss*, *Ranking Loss*, *Average Precision*, *Coverage* and *OneError*. We set the number of nearest neighbors K to 10 and the smoothing factor to 1. Additionally, all code has been released in the open source GITHUB library at <https://github.com/huangweiyi56/SRLG-LMA>.

6.3. Experimental analysis

In this section, we analyze the performance of the proposed algorithm SRLG-LMA and the difference between the proposed algorithm SRLG-LMA and the selected comparison algorithms from three aspects: ablation studies, performance analysis and statistical analysis.

6.3.1. Ablation studies

Strongly relevant label gain (*SRLG*) and label mutual aid (*MINT*) play a crucial role in our work. Here, we demonstrate the effectiveness of the strongly relevant label gain and label mutual aid through ablation studies [38]. In existing studies [2,14], maximum relevance and minimum redundancy are considered together. Therefore, our work is composed of three different modules, including maximum relevant and minimum redundancy (*MREL* + *MRED*, shortly *MRMR*), Strongly relevant label gain (*SRLG*), and label mutual aid (*MINT*). Tables 7–8 show five evaluation metrics for the Emotions and the Birds datasets.

In Table 7, The highest performance is obtained when all the modules are available while the performance is lower when only the

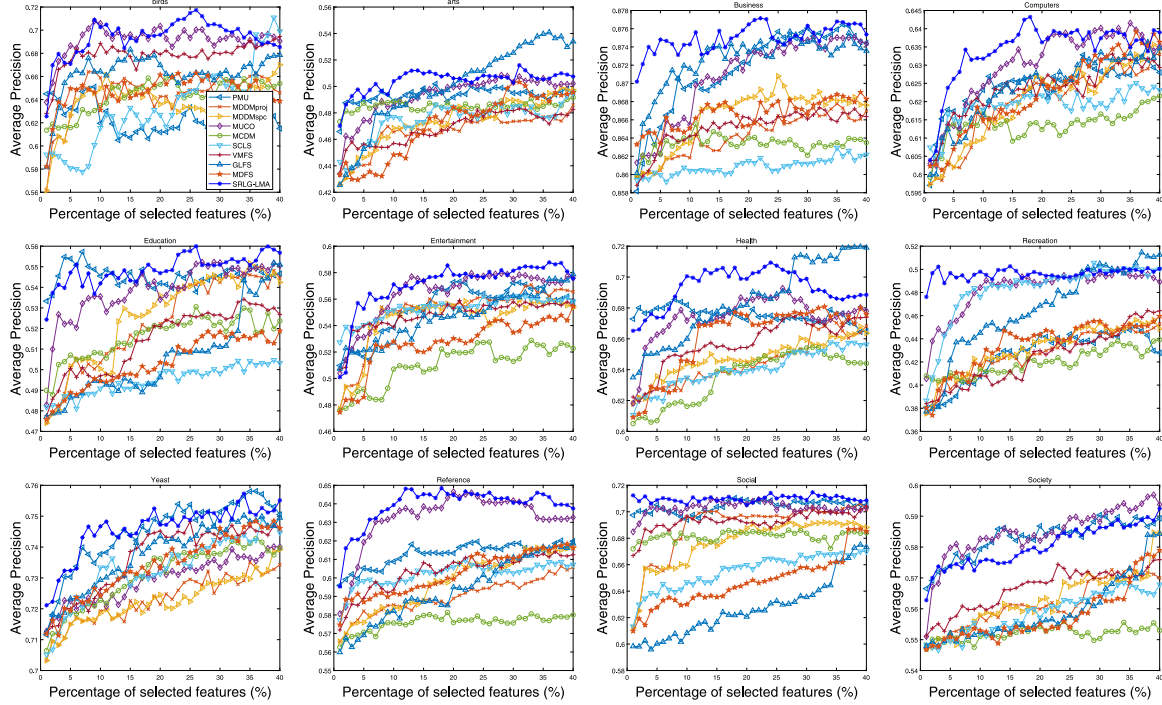


Fig. 6. The change of the Average Precision metric of different algorithms as the selected feature increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7
Ablation analysis based on the Emotions dataset.

MRMR	✓	✓	✓	✓
SRLG	×	✓	×	✓
MINT	×	×	✓	✓
AP	0.7240	0.7295	0.7294	0.7322
HL	0.2628	0.2657	0.2621	0.2612
OE	0.3902	0.3919	0.3927	0.3890
CV	2.2835	2.2335	2.2318	2.2185
RL	0.2369	0.2307	0.2315	0.2289

Table 8
Ablation analysis based on the Birds dataset.

MRMR	✓	✓	✓	✓
SRLG	×	✓	×	✓
MINT	×	×	✓	✓
AP	0.6831	0.6901	0.6924	0.6932
HL	0.0535	0.0539	0.0528	0.0528
OE	0.3752	0.3678	0.3741	0.3734
CV	3.7529	3.6663	3.5192	3.5076
RL	0.1399	0.1354	0.1314	0.1308

module *MRMR*. When *SRLG* and *MINT* are combined with *MRMR* respectively, their performance is mostly higher than that of single module *MRMR*. And when the three modules are combined, the five evaluation metrics achieve the highest level. According to Table 8, even though the performance of the three modules combined together is not the best in terms of the *OE* metric, it is still sub-optimal, and the best one also includes strongly relevant label gain. From a global perspective, when the three modules work together, their performance is the best. Clearly, we can verify the effectiveness of Strongly relevant label gain (*SRLG*) and label mutual aid (*MINT*) by ablation analysis. Furthermore, we illustrate the superiority of the proposed algorithm through performance analysis.

6.3.2. Performance analysis

By comparing the five evaluation metrics among SRLG-LMA, PMU, MDDMProj, MDDMSpc, MUCO, MCDM, SCLS, VMFS, GLFS and MDFS, we can verify the superiority of the proposed algorithm. The specific experimental results are shown in figures and tables.

Given that SRLG-LMA and other comparison algorithms can generate a sequence of features, we select $NumK = 40\%$, representing 40% of the total number of features in the datasets, as the results of the experimental comparison. Wherein, Tables 9–13 present the average values of the evaluation metrics for the range of 1%–40% of the total number of features. Figs. 6–10 illustrate the performance of the proposed algorithm and other comparison algorithms for the range of 1% to 40%, with a step size of 1%. The symbol ↓ indicates better performance when the value in the table is smaller, while the ↑ symbol represents better performance when the value in the table is larger. The bold in the table represents the optimal result from a single dataset. The Average is used to represent the average result of a single algorithm across all benchmark datasets. The Rank.Avg represents the average ranking of any algorithm across all benchmark datasets, with smaller values indicating superior performance. Table 9 displays the *AP* of all comparison algorithms under benchmark datasets. Table 10 shows the performance of all algorithms under the *HL* metric. Table 11 exhibits the performance of all algorithms under the *OE*. Table 12 represents the *RL* of all algorithms under benchmark datasets. Table 13 illustrates the performance of all algorithms under the *CV*.

Regarding the *AP* metric in Table 9, it is apparent that the SRLG-LMA algorithm exhibits superior performance on twelve benchmark datasets. Additionally, the MUCO algorithm outperforms other algorithms on two datasets. The MCDM and PMU algorithms are leading on one dataset. Nevertheless, the SRLG-LMA achieves better rankings on four additional benchmark datasets, such as 3rd on Emotions and Society, and 2nd on Flags. In terms of Average, the *AP* metric of the SRLG-LMA algorithm shows the highest value. According to

Table 9

Performance comparison results of the feature sets selected by different algorithms on AP metric (↑).

Data set	PMU [11]	MDDMProj [34]	MDDMSpc [34]	MUCO [26]	MCDM [35]	SCLS [25]	VMFS [36]	GLFS [24]	MDFS [22]	SRLG-LMA
Emotions	0.6628	0.7323	0.7251	0.7294	0.7336	0.6273	0.6244	0.7384	0.6088	0.7322
Birds	0.6244	0.6451	0.6381	0.6906	0.6423	0.6316	0.6766	0.6615	0.6448	0.6932
Yeast	0.7442	0.7227	0.7223	0.7290	0.7305	0.7324	0.7360	0.7370	0.7334	0.7453
Arts	0.4930	0.4656	0.4721	0.4917	0.4845	0.4731	0.4678	0.4988	0.4701	0.5034
Business	0.8715	0.8642	0.8663	0.8707	0.8632	0.8608	0.8651	0.8725	0.8666	0.8749
Computers	0.6238	0.6193	0.6194	0.6294	0.6133	0.6187	0.6230	0.6232	0.6219	0.6337
Education	0.5449	0.5219	0.5252	0.5376	0.5160	0.4946	0.5135	0.5049	0.5031	0.5485
Health	0.6749	0.6443	0.6473	0.6730	0.6353	0.6400	0.6561	0.6835	0.6603	0.6927
Recreation	0.4255	0.4256	0.4308	0.4811	0.4193	0.4806	0.4228	0.4640	0.4339	0.4948
Reference	0.6123	0.5897	0.5973	0.6317	0.5758	0.6014	0.6028	0.5933	0.6018	0.6383
Society	0.5826	0.5612	0.5626	0.5838	0.5520	0.5585	0.5660	0.5587	0.5573	0.5799
Entertainment	0.5517	0.5485	0.5431	0.5631	0.5098	0.5521	0.5473	0.5458	0.5267	0.5709
Social	0.7038	0.6879	0.6767	0.7026	0.6815	0.6545	0.6922	0.6263	0.6457	0.7094
Medical	0.6850	0.4984	0.4975	0.4816	0.5911	0.7312	0.4753	0.3985	0.4750	0.7361
Flags	0.7889	0.7969	0.7926	0.8098	0.7961	0.7956	0.7982	0.7847	0.7962	0.8001
Scene	0.8041	0.7658	0.7479	0.7336	0.7731	0.7539	0.6926	0.7626	0.7478	0.7518
Average	0.6496	0.6306	0.6290	0.6462	0.6323	0.6379	0.6225	0.6284	0.6183	0.6691
Rank.Avg	4.0625	6.2500	6.8125	3.6875	7.1250	6.8125	6.1250	5.3750	7.0625	1.6875

Table 10

Performance comparison results of the feature sets selected by different algorithms on HL metric (↓).

Data set	PMU [11]	MDDMProj [34]	MDDMSpc [34]	MUCO [26]	MCDM [35]	SCLS [25]	VMFS [36]	GLFS [24]	MDFS [22]	SRLG-LMA
Emotions	0.3045	0.2645	0.2697	0.2618	0.2638	0.3197	0.3183	0.2482	0.3335	0.2612
Birds	0.0686	0.0647	0.0652	0.0539	0.0624	0.0671	0.0552	0.0620	0.0617	0.0528
Yeast	0.2100	0.2204	0.2213	0.2147	0.2163	0.2161	0.2116	0.2099	0.2132	0.2055
Arts	0.0612	0.0626	0.0624	0.0618	0.0598	0.0613	0.0627	0.0601	0.0624	0.0609
Business	0.0279	0.0286	0.0286	0.0280	0.0282	0.0287	0.0286	0.0280	0.0285	0.0275
Computers	0.0401	0.0416	0.0408	0.0401	0.0411	0.0409	0.0408	0.0409	0.0407	0.0396
Education	0.0412	0.0428	0.0426	0.0418	0.0409	0.0442	0.0434	0.0434	0.0430	0.0419
Health	0.0433	0.0468	0.0472	0.0437	0.0468	0.0483	0.0454	0.0446	0.0461	0.0418
Recreation	0.0641	0.0640	0.0637	0.0609	0.0627	0.0599	0.0639	0.0611	0.0636	0.0603
Reference	0.0305	0.0339	0.0337	0.0297	0.0300	0.0301	0.0307	0.0338	0.0319	0.0295
Society	0.0570	0.0591	0.0590	0.0573	0.0590	0.0577	0.0578	0.0588	0.0592	0.0573
Entertainment	0.0621	0.0624	0.0632	0.0612	0.0631	0.0630	0.0627	0.0626	0.0632	0.0603
Social	0.0255	0.0263	0.0276	0.0256	0.0243	0.0282	0.0255	0.0316	0.0302	0.0251
Medical	0.0199	0.0258	0.0249	0.0265	0.0197	0.0187	0.0262	0.0276	0.0276	0.0190
Flags	0.3027	0.3173	0.3291	0.2934	0.2948	0.3236	0.2984	0.3324	0.3143	0.2808
Scene	0.1137	0.1270	0.1320	0.1441	0.1260	0.1329	0.1485	0.1331	0.1330	0.1396
Average	0.0920	0.0930	0.0944	0.0903	0.0899	0.0963	0.0950	0.0924	0.0970	0.0877
Rank.Avg	4.0625	7.1250	7.2500	3.9375	4.6250	6.6250	6.4375	5.8750	7.0000	2.0625

Rank.Avg, the SRLG-LMA algorithm achieves the highest ranking. Evidently, the SRLG-LMA algorithm significantly outperforms the other nine algorithms on AP metric.

The HL metric results in Table 10 clearly indicate that the SRLG-LMA algorithm outperforms the others on eight benchmark datasets, while the MCDM algorithm excels on Arts, Education and Social. Additionally, the SCLS demonstrates superior performance on Recreation and Medical, while GLFS excels in Emotions. Furthermore, considering the Average, the SRLG-LMA algorithm shows a HL value that is 0.0026 lower than the suboptimal MUCO algorithm and 0.0093 lower than the HL value generated by the worst-performing MDFS algorithm. The Rank.Avg demonstrates that the SRLG-LMA algorithm holds the highest position, clearly indicating its superiority over other algorithms in terms of the HL metric.

Table 11 presents the performance of various algorithms based on the OE metric. Among them, the SRLG-LMA algorithm demonstrates the highest performance across ten benchmark datasets. Additionally, the PMU algorithm shows excellent performance on the Yeast and Society datasets, while the MUCO, GLFS, and SCLS algorithms exhibit the best performance on the Flags, Emotions and Medical benchmark datasets, respectively. In terms of the Average, the SRLG-LMA shows a significant gap compared to other algorithms in OE values. In particular, it displays a 0.0246 difference with the suboptimal algorithm in OE values. Moreover, when considering the Rank.Avg in the last row, the SRLG-LMA algorithm ranks the highest, providing further evidence of its superiority over other algorithms on the OE metric.

The SRLG-LMA algorithm outperforms all other algorithms on nine benchmark datasets, as indicated by the RL metric in Table 12. The MUCO algorithm achieves the best performance on the Society, Flags and Birds datasets. The PMU algorithm shows superior performance on the Yeast dataset, while the MDDMProj algorithm outperforms others on the Emotions dataset. The GLFS algorithm exhibits outstanding performance on the Business dataset. The SRLG-LMA algorithm has the lowest values in the Average, indicating its superior overall performance. The SRLG-LMA also achieves the highest rank on Rank.Avg. The results clearly demonstrate that the SRLG-LMA algorithm surpasses the other nine algorithms in terms of the RL metric.

Table 13 presents the results obtained from the CV metric. The SRLG-LMA algorithm demonstrates the highest performance on nine benchmark datasets. The MUCO algorithm achieves the best performance on the Birds and Society datasets. The PMU, GLFS, MDFS and MDDMProj algorithms demonstrate outstanding performance on the Yeast, Business, Flags and Emotions datasets, respectively. SRLG-LMA obtains the lowest value on both Average and Rank.Avg, indicating superior performance compared to other representative comparison algorithms on the CV metric.

Based on the performance analysis conducted, the proposed algorithm SRLG-LMA demonstrates superior performance compared to other algorithms in the majority of cases. The aforementioned five widely used performance metrics indicate that SRLG-LMA exhibits better classification performance. To further validate the accuracy and reliability of the algorithms, we present the performance trends of each algorithm under varying datasets based on the number of selected

Table 11

Performance comparison results of the feature sets selected by different algorithms on OE metric (↓).

Data set	PMU [11]	MDDMPProj [34]	MDDMSpc [34]	MUCO [26]	MCDM [35]	SCLS [25]	VMFS [36]	GLFS [24]	MDFS [22]	SRLG-LMA
Emotions	0.4612	0.3831	0.3899	0.3897	0.3803	0.4959	0.5012	0.3771	0.5215	0.3890
Birds	0.5186	0.4736	0.4842	0.3807	0.4568	0.4880	0.4004	0.4553	0.4628	0.3735
Yeast	0.2379	0.2520	0.2530	0.2533	0.2557	0.2509	0.2512	0.2495	0.2503	0.2448
Arts	0.6497	0.6983	0.6892	0.6570	0.6547	0.6843	0.6949	0.6481	0.6898	0.6396
Business	0.1278	0.1351	0.1343	0.1284	0.1337	0.1368	0.1359	0.1289	0.1342	0.1230
Computers	0.4526	0.4598	0.4590	0.4481	0.4708	0.4564	0.4552	0.4535	0.4528	0.4447
Education	0.5918	0.6300	0.6218	0.6041	0.6324	0.6666	0.6392	0.6457	0.6559	0.5911
Health	0.4216	0.4544	0.4510	0.4231	0.4647	0.4618	0.4463	0.4092	0.4350	0.3972
Recreation	0.7420	0.7448	0.7340	0.6667	0.7445	0.6609	0.7455	0.6854	0.7311	0.6493
Reference	0.4904	0.5093	0.5039	0.4631	0.5248	0.4873	0.5019	0.5039	0.4910	0.4559
Society	0.4648	0.4891	0.4872	0.4650	0.5003	0.4864	0.4857	0.4914	0.4944	0.4698
Entertainment	0.6044	0.6074	0.6172	0.5849	0.6470	0.6121	0.6109	0.6187	0.6401	0.5727
Social	0.3870	0.4111	0.4302	0.3911	0.4180	0.4633	0.4023	0.5215	0.4847	0.3810
Medical	0.3977	0.6191	0.6144	0.6477	0.4767	0.3366	0.6502	0.7316	0.6599	0.3373
Flags	0.2423	0.2442	0.2404	0.2058	0.2288	0.2327	0.2269	0.2481	0.2231	0.2308
Scene	0.3126	0.3679	0.3938	0.4328	0.3626	0.3934	0.4749	0.3887	0.3982	0.4090
Average	0.4439	0.4675	0.4690	0.4463	0.4595	0.4571	0.4764	0.4723	0.4828	0.4193
Rank.Avg	3.7500	6.8125	6.6250	3.8750	6.6250	6.2500	6.5625	5.6875	6.6875	2.1250

Table 12

Performance comparison results of the feature sets selected by different algorithms on RL metric (↓).

Data set	PMU [11]	MDDMPProj [34]	MDDMSpc [34]	MUCO [26]	MCDM [35]	SCLS [25]	VMFS [36]	GLFS [24]	MDFS [22]	SRLG-LMA
Emotions	0.3299	0.2246	0.2408	0.2308	0.2316	0.3752	0.3810	0.2247	0.4042	0.2290
Birds	0.1364	0.1428	0.1493	0.1287	0.1530	0.1580	0.1373	0.1328	0.1464	0.1309
Yeast	0.1828	0.2002	0.1996	0.1959	0.1930	0.1906	0.1886	0.1889	0.1901	0.1835
Arts	0.1559	0.1629	0.1614	0.1535	0.1665	0.1621	0.1611	0.1574	0.1634	0.1505
Business	0.0430	0.0456	0.0446	0.0436	0.0473	0.0483	0.0454	0.0409	0.0444	0.0420
Computers	0.0957	0.0951	0.0967	0.0932	0.0973	0.0995	0.0961	0.0951	0.0969	0.0924
Education	0.0932	0.0974	0.0971	0.0943	0.1011	0.1036	0.0997	0.1034	0.1030	0.0918
Health	0.0637	0.0712	0.0706	0.0640	0.0758	0.0745	0.0678	0.0642	0.0689	0.0603
Recreation	0.1994	0.2005	0.2012	0.1832	0.2095	0.1902	0.2022	0.1922	0.1999	0.1784
Reference	0.0888	0.0963	0.0935	0.0840	0.0997	0.0960	0.0898	0.0975	0.0944	0.0823
Society	0.1469	0.1546	0.1535	0.1463	0.1591	0.1575	0.1513	0.1569	0.1560	0.1476
Entertainment	0.1274	0.1299	0.1306	0.1243	0.1444	0.1284	0.1287	0.1273	0.1351	0.1221
Social	0.0673	0.0702	0.0717	0.0662	0.0751	0.0759	0.0703	0.0755	0.0752	0.0653
Medical	0.0778	0.1147	0.1181	0.1133	0.1019	0.0675	0.1171	0.1400	0.1121	0.0608
Flags	0.2472	0.2384	0.2419	0.2308	0.2430	0.2421	0.2412	0.2457	0.2329	0.2375
Scene	0.1260	0.1577	0.1733	0.1666	0.1481	0.1601	0.2165	0.1473	0.1679	0.1510
Average	0.1363	0.1376	0.1402	0.1324	0.1404	0.1456	0.1496	0.1369	0.1494	0.1266
Rank.Avg	3.6250	6.1250	6.8125	3.1250	8.0000	7.4375	6.0625	5.2500	6.8125	1.7500

Table 13

Performance comparison results of the feature sets selected by different algorithms on CV metric (↓).

Data set	PMU [11]	MDDMPProj [34]	MDDMSpc [34]	MUCO [26]	MCDM [35]	SCLS [25]	VMFS [36]	GLFS [24]	MDFS [22]	SRLG-LMA
Emotions	2.7124	2.1939	2.2772	2.2533	2.2375	2.9814	3.0089	2.1992	3.1173	2.2185
Birds	3.5267	3.7095	3.8522	3.4506	4.0288	4.0910	3.6314	3.5407	3.8938	3.5090
Yeast	6.5587	6.7897	6.7575	6.7331	6.7085	6.6231	6.6460	6.6210	6.6210	6.5741
Arts	5.5719	5.7263	5.7040	5.4857	5.8540	5.7360	5.6728	5.6165	5.7555	5.4328
Business	2.3893	2.4609	2.4321	2.3886	2.5490	2.5686	2.4618	2.3034	2.4495	2.3447
Computers	4.5649	4.5497	4.6043	4.4788	4.6204	4.7156	4.5858	4.5318	4.6040	4.4344
Education	3.9610	4.0991	4.0948	4.0030	4.2160	4.3221	4.1869	4.3278	4.2970	3.9122
Health	3.4128	3.6878	3.6652	3.4270	3.8544	3.8153	3.5743	3.4277	3.6141	3.2832
Recreation	5.2380	5.2543	5.2666	4.9112	5.4401	5.0649	5.2791	5.0862	5.2479	4.8079
Reference	3.4270	3.6809	3.5930	3.2690	3.8008	3.6791	3.4638	3.7235	3.6304	3.2170
Society	5.7788	5.9881	5.9657	5.7450	6.1187	6.1000	5.9022	6.0623	6.0464	5.7875
Entertainment	3.4012	3.4413	3.4545	3.3411	3.7755	3.4116	3.4232	3.3805	3.5527	3.2977
Social	3.5613	3.6864	3.7604	3.5156	3.9114	3.9180	3.6809	3.8985	3.8926	3.4696
Medical	4.3712	6.0175	6.1862	5.9535	5.4845	3.8467	6.1280	7.0516	5.8742	3.5811
Flags	3.9327	3.8769	3.9135	3.8481	3.8423	3.9635	3.9135	3.9673	3.8231	3.8288
Scene	0.7337	0.8920	0.9715	0.9398	0.8407	0.9048	1.1887	0.8405	0.9454	0.8622
Average	3.9463	4.1284	4.1562	3.9840	4.2052	4.1118	4.1703	4.1627	4.2103	3.7850
Rank.Avg	3.5000	6.1250	6.7500	3.3125	7.7500	7.5625	6.2500	5.4375	6.6250	1.6875

features. We take the first 40% of the sorted set obtained by feature selection as the process of shift, and its feature enhancement step is set to 1%. The horizontal axis in the figure represents the size of the feature subset, the vertical axis represents the performance corresponding to that feature subset, and the different colors indicate the performance trends in different algorithms.

The changing trends of the comparison methods under different evaluation metrics are shown in Figs. 6–10. Fig. 6 presents the results of the performance of different algorithms in terms of *AP*. Fig. 7 displays the experimental results of the *CV* metric. Fig. 8 illustrates the *HL* metric. The result from the *OE* metric is depicted in Fig. 9. Fig. 10 exhibits the results of the *RL* metric under different algorithms.

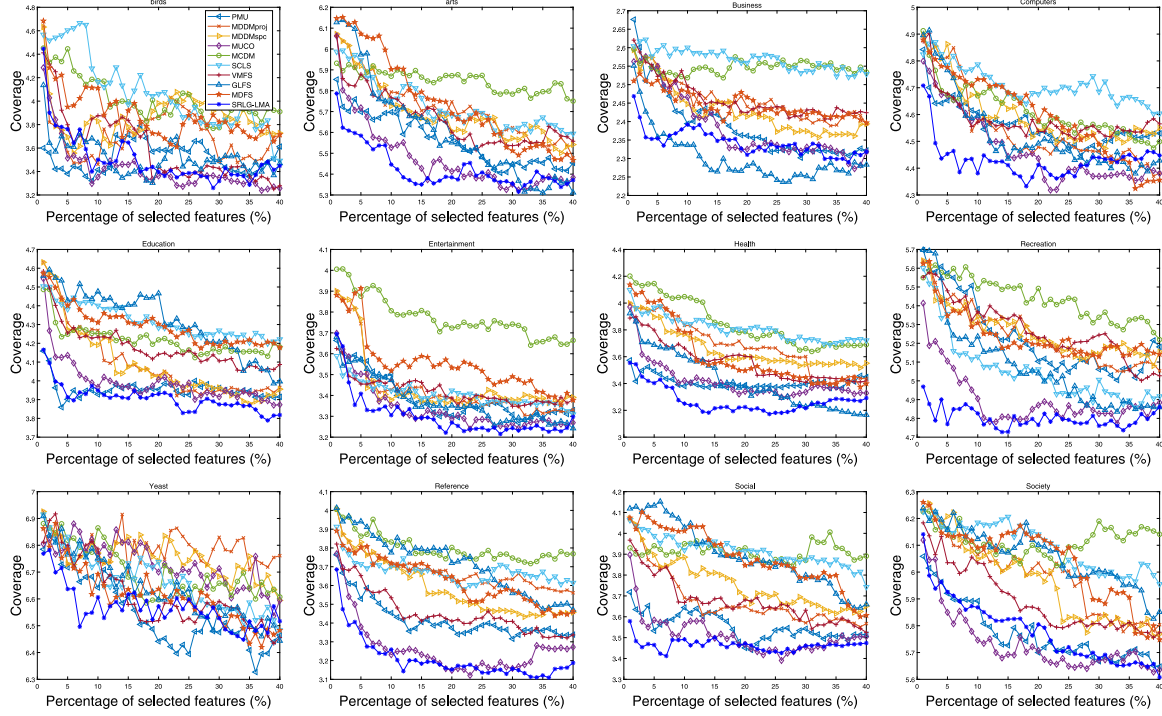


Fig. 7. The change of the Coverage metric of different algorithms as the selected feature increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

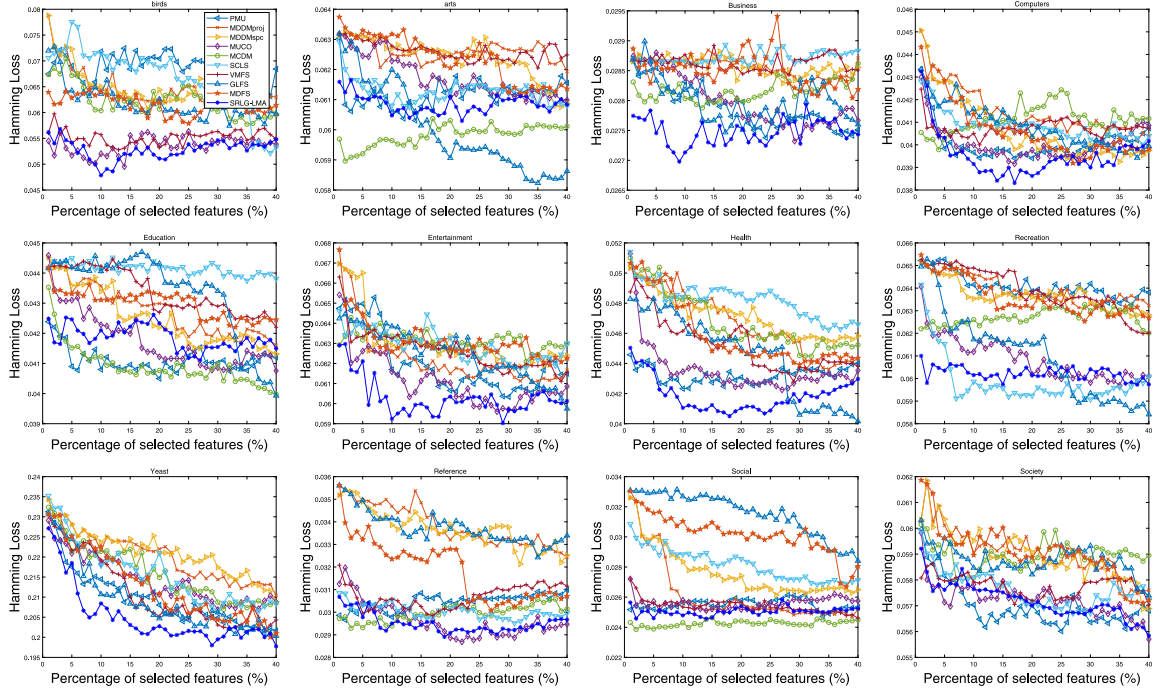


Fig. 8. The change of the Hamming Loss metric of different algorithms as the selected feature increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

From Fig. 6, the results demonstrate that SRLG-LMA outperforms other algorithms on these twelve real-world benchmark datasets. Furthermore, as the number of features increases, it basically demonstrates superiority to other methods, indicating the stability of SRLG-LMA. The figure clearly illustrates that both the maximum and average values of the *AP* metric for SRLG-LMA surpass those of other algorithms on

twelve real-world benchmark datasets. Consequently, we conclude that the SRLG-LMA algorithms outperform other comparison algorithms.

Figs. 7–10 show the superiority over SRLG-LMA under the remaining four metrics, and it can be seen that the SRLG-LMA algorithm has obvious advantages over other algorithms. Apparently, the SRLG-LMA performs adequately in the selected benchmark datasets and is more

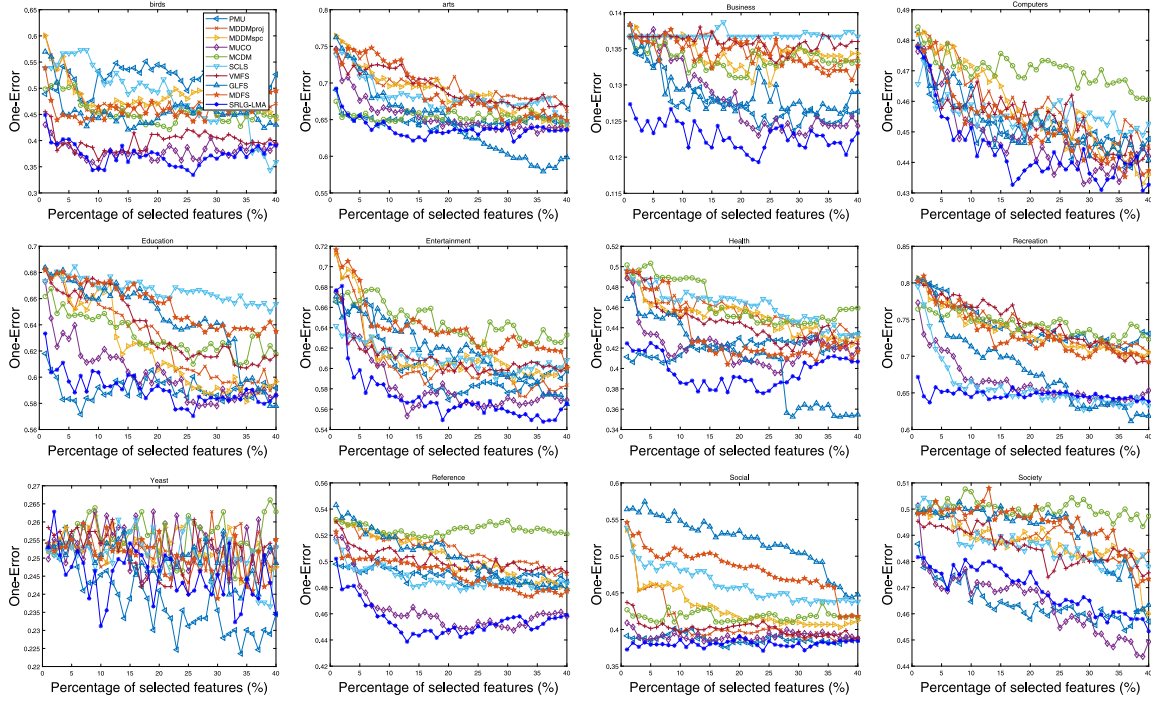


Fig. 9. The change of the One Error metric of different algorithms as the selected feature increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

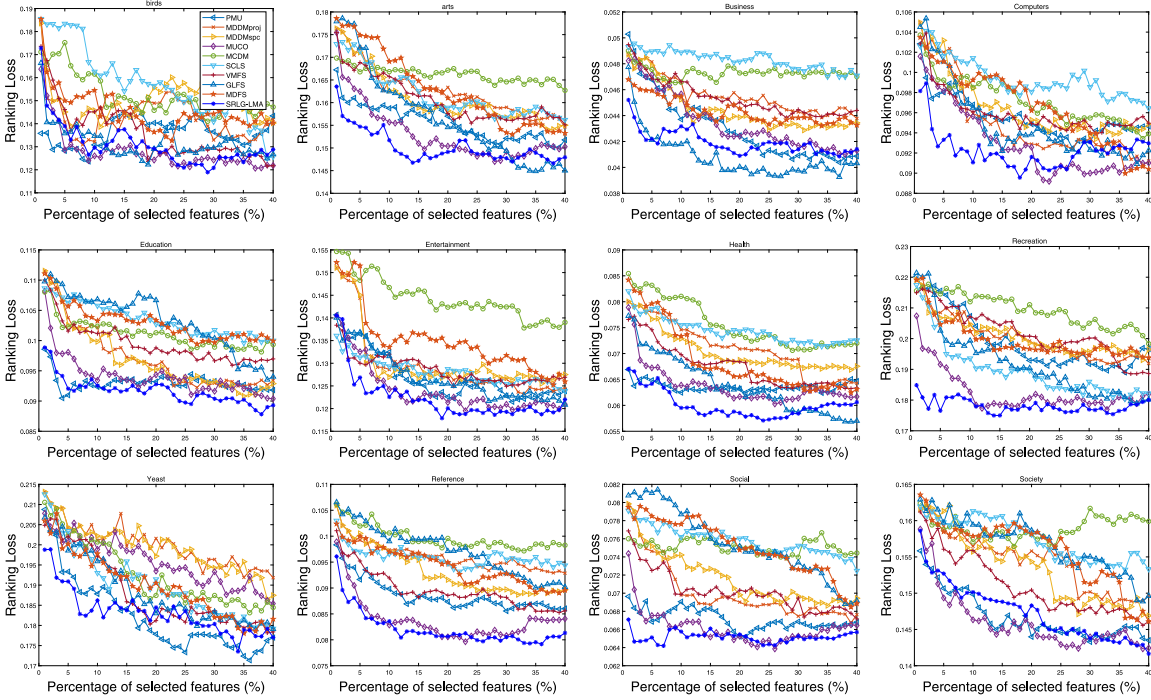


Fig. 10. The change of the Ranking Loss metric of different algorithms as the selected feature increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

stable, which cannot be achieved by other representative comparison algorithms.

6.3.3. Statistical analysis

To better compare the performance difference between the proposed algorithm and the other algorithms, the statistical significance of the experimental results is investigated in this section using the Friedman test [39] and the Bonferroni–Dunn test [40].

The Friedman test is used to compare the overall performance of k algorithms on N benchmark datasets. Given r_i^p to represent the i th algorithm's ranking in the p th dataset, $R_i = \frac{1}{N} \sum_{p=1}^N r_i^p$ represents the average ranking of the i th algorithm across all benchmark datasets. Therefore, the Chi-Square definition is as follows:

$$\chi_F^2 = \frac{12 * N}{k * (k + 1)} * \left(\sum_{i=1}^k R_i^2 - \frac{k * (k + 1)^2}{4} \right), \quad (21)$$

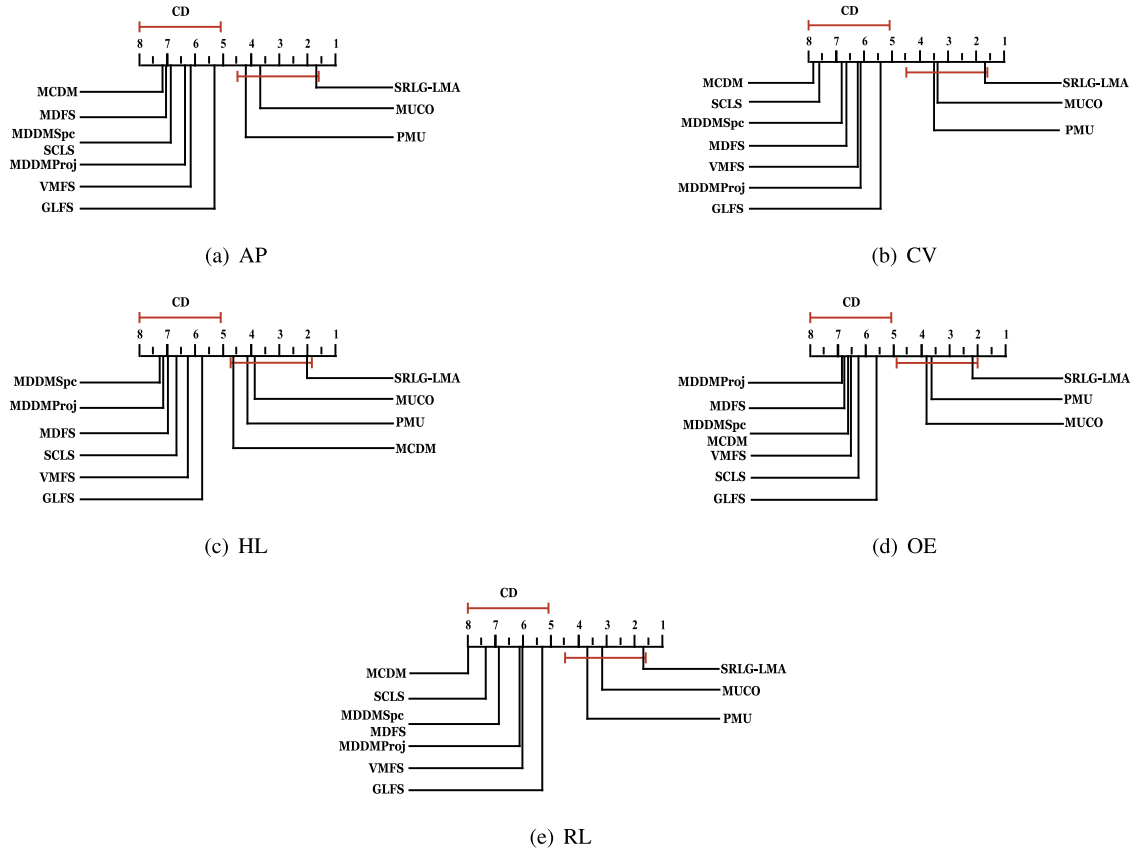


Fig. 11. The CD plots of different evaluation metrics. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 14
Friedman test under different evaluation metrics.

Evaluation metrics	F_F	Critical value
Average Precision	8.2975	
Coverage	11.8566	
Hamming Loss	7.4299	1.9499
One Error	6.3305	
Ranking Loss	12.4619	

where the degree of freedom about Chi-Square χ_F^2 is $(k-1)$. Then under the null hypothesis, that all experimental algorithms are equivalent, the Friedman statistical formula for Carmen is given as follows:

$$F_F = \frac{(N-1) * \chi_F^2}{N(k-1) - \chi_F^2}. \quad (22)$$

The F_F satisfies the F -distribution with degrees of freedom $(k-1)$ and $(k-1)(N-1)$. Evidently, Table 14 displays Friedman statistics for different metrics of MLKNN. The $k=10$ and $N=16$, significance level $\alpha=0.05$. Therefore, it is possible to obtain a Critical value of 2.1044 from code (`icdf('F', 1 -  $\alpha$ ,  $k-1$ ,  $(k-1) * (N-1)$ )`).

As demonstrated in Table 14, we can reject the null hypothesis based on the Friedman test statistics. This is because the test results of the five evaluation metrics are considerably higher than 1.9499. In other words, there is a substantial difference in performance among all the algorithms. To further validate these findings, the Bonferroni-Dunn test can be conducted to effectively illustrate the performance gap between the proposed algorithm and the other comparison algorithms.

In general, using the Critical Difference as a baseline for the Bonferroni-Dunn test, the formula for the Critical Difference (shortly CD) is as follows:

$$CD_\alpha = \rho_\alpha \sqrt{\frac{k * (k+1)}{6 * N}}, \quad (23)$$

where N and k are defined as mentioned before. According to the cut-off value query table in the Bonferroni-Dunn test, $\rho_\alpha = 2.773$ can be obtained at $k=10$, $\alpha=0.05$. Therefore, at $N=16$, we can compute $CD_\alpha = 2.9683$.

In Fig. 11, the CD plots of various evaluation metrics are presented. These plots illustrate the average rankings of the experimental algorithms, plotted along the coordinate axis. The algorithms with better performance are depicted on the right side of the axis. In order to assess the performance of the SRLG-LMA algorithm, it can be treated as the control algorithm. Additionally, the red CD line marks the interval between SRLG-LMA and all other comparison algorithms. Then, if an algorithm falls outside of this red line region, its performance is deemed significantly different from that of the control algorithm. Conversely, if an algorithm falls within this red line region, its performance is considered less dissimilar to that of the control algorithm.

Based on Fig. 11, it can be concluded that on AP , CV , OE and RL , the SRLG-LMA performs significantly better than MCDM, MDDMproj, MDDMSpc, SCLS, VMFS, GLFS and MDFS. Moreover, the SRLG-LMA performs significantly better than MDDMproj, MDDMSpc, SCLS, VMFS, GLFS and MDFS on HL . In addition, even though MUCO and PMU are both within the CD range of SRLG-LMA, we can see that there is still a large performance gap with SRLG-LMA. Furthermore, the SRLG-LMA consistently maintains the top ranking. In conclusion, the SRLG-LMA significantly outperforms the other comparison algorithms across all five evaluation metrics.

In summary, combined with the above discussions, it is evident that SRLG-LMA achieves better classification performance in most cases than the compared representative multi-label feature selection algorithms. This result aligns with our previous theoretical analysis and corroborates the effectiveness of considering strongly relevant label gain and label mutual aid into the multi-label feature selection process.

7. Conclusions and future work

This paper presents a novel method for multi-label feature selection based on label mutual aid and strongly relevant label gain (SRLG-LMA). To investigate potential correlations between labels in greater depth, we construct the mutual aid information between labels based on fuzzy conditional mutual information, which has not been adequately considered in previous studies based on information theory. For strongly relevant label gain, it additionally exploits the gain between candidate features and labels relative to selected features, and captures the label subset with the highest degree of relevance to candidate features. This allows us to further consider positive correlations between features, which is neglected by the existing multi-label feature selection methods.

To verify the superiority of the proposed algorithm, we conduct extensive experiments on sixteen benchmark multi-label datasets. Firstly, we demonstrate the effectiveness of label mutual aid and strongly relevant label gain through ablation studies. Secondly, we compare the performance among the proposed algorithm and nine representative multi-label feature selection algorithms (PMU, MDDMProj, MDDMSpc, MUCO, MCDM, SCLS, VMFS, GLFS and MDFS). Finally, the experimental results demonstrate that the proposed algorithm in this study achieves exceptional classification performance.

Although the proposed algorithm can reduce the dimensionality effectively, it considers only complete multi-label data, i.e. multi-label data without missing feature values. Actually, incomplete multi-label data, i.e. multi-label data contain missing feature values exist widely in many applications. Meanwhile, the proposed algorithm is not efficient in handling multi-label data with noise, which may lead to inaccurate results of classification. Therefore, our future work will focus on further investigating feature selection methods for incomplete multi-label data and multi-label data with noise.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

Data availability

Data will be made available on request.

Acknowledgments

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References

- [1] J. Dai, Q. Hu, J. Zhang, H. Hu, N. Zheng, Attribute selection for partially labeled categorical data by rough set approach, *IEEE Trans. Cybern.* 47 (9) (2017) 2460–2471.
- [2] H. Peng, F. Long, C. Ding, Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy, *IEEE Trans. Pattern Anal. Mach. Intell.* 27 (8) (2005) 1226–1238.
- [3] Y. Chai, H. Liu, J. Xu, S. Samtani, Y. Jiang, H. Liu, A multi-label classification with an adversarial-based denoising autoencoder for medical image annotation, *ACM Trans. Manage. Inf. Syst.* 14 (2) (2023) 1–21.
- [4] X. Li, H. Wu, M. Li, H. Liu, Multi-label video classification via coupling attentional multiple instance learning with label relation graph, *Pattern Recognit. Lett.* 156 (2022) 53–59.
- [5] Y. Ma, X. Liu, L. Zhao, Y. Liang, P. Zhang, B. Jin, Hybrid embedding-based text representation for hierarchical multi-label text classification, *Expert Syst. Appl.* 187 (2022) 115905.
- [6] R. Huang, W. Jiang, G. Sun, Manifold-based constraint laplacian score for multi-label feature selection, *Pattern Recognit. Lett.* 112 (2018) 346–352.
- [7] J. Dai, J. Chen, Y. Liu, H. Hu, Novel multi-label feature selection via label symmetric uncertainty correlation learning and feature redundancy evaluation, *Knowl.-Based Syst.* 207 (2020) 106342.
- [8] R.B. Pereira, A. Plastino, B. Zadrozny, L.H. Merschmann, Categorizing feature selection methods for multi-label classification, *Artif. Intell. Rev.* 49 (2018) 57–78.
- [9] J. Lee, D.-W. Kim, Efficient multi-label feature selection using entropy-based label selection, *Entropy* 18 (11) (2016) 405.
- [10] P. Zhu, Q. Xu, Q. Hu, C. Zhang, H. Zhao, Multi-label feature selection with missing labels, *Pattern Recognit.* 74 (2018) 488–502.
- [11] J. Lee, D.-W. Kim, Feature selection for multi-label classification using multivariate mutual information, *Pattern Recognit. Lett.* 34 (3) (2013) 349–357.
- [12] J. Zhang, Y. Lin, M. Jiang, S. Li, Y. Tang, K.C. Tan, Multi-label feature selection via global relevance and redundancy optimization, in: 29th International Joint Conference on Artificial Intelligence (IJCAI 2020), International Joint Conferences on Artificial Intelligence, 2021, pp. 2512–2518.
- [13] J. Lee, D.-W. Kim, Fast multi-label feature selection based on information-theoretic feature ranking, *Pattern Recognit.* 48 (9) (2015) 2761–2771.
- [14] F. Li, D. Miao, W. Pedrycz, Granular multi-label feature selection based on mutual information, *Pattern Recognit.* 67 (2017) 410–423.
- [15] H. Lim, J. Lee, D.-W. Kim, Optimization approach for feature selection in multi-label classification, *Pattern Recognit. Lett.* 89 (2017) 25–30.
- [16] J. Wan, H. Chen, Z. Yuan, T. Li, X. Yang, B. Sang, A novel hybrid feature selection method considering feature interaction in neighborhood rough set, *Knowl.-Based Syst.* 227 (2021) 107167.
- [17] W. Qian, X. Long, Y. Wang, Y. Xie, Multi-label feature selection based on label distribution and feature complementarity, *Appl. Soft Comput.* 90 (2020) 106167.
- [18] J. Read, A pruned problem transformation method for multi-label classification, in: *Proc. 2008 New Zealand Computer Science Research Student Conference (NZCSRS 2008)*, Vol. 143150, 2008, p. 41.
- [19] M.R. Boutell, J. Luo, X. Shen, C.M. Brown, Learning multi-label scene classification, *Pattern Recognit.* 37 (9) (2004) 1757–1771.
- [20] G. Tsoumakas, I. Vlahavas, Random k-labelsets: An ensemble method for multilabel classification, in: *Machine Learning: ECML 2007: 18th European Conference on Machine Learning*, Warsaw, Poland, September 17–21, 2007. *Proceedings 18*, Springer, 2007, pp. 406–417.
- [21] J. Huang, G. Li, Q. Huang, X. Wu, Joint feature selection and classification for multilabel learning, *IEEE Trans. Cybern.* 48 (3) (2018) 876–889.
- [22] J. Zhang, Z. Luo, C. Li, C. Zhou, S. Li, Manifold regularized discriminative feature selection for multi-label learning, *Pattern Recognit.* 95 (2019) 136–150.
- [23] H. Lim, D.-W. Kim, Mfc: Initialization method for multi-label feature selection based on conditional mutual information, *Neurocomputing* 382 (2020) 40–51.
- [24] J. Zhang, H. Wu, M. Jiang, J. Liu, S. Li, Y. Tang, J. Long, Group-preserving label-specific feature selection for multi-label learning, *Expert Syst. Appl.* 213 (2023) 118861.
- [25] J. Lee, D.-W. Kim, Scis: Multi-label feature selection based on scalable criterion for large label set, *Pattern Recognit.* 66 (2017) 342–352.
- [26] Y. Lin, Q. Hu, J. Liu, J. Li, X. Wu, Streaming feature selection for multilabel learning based on fuzzy mutual information, *IEEE Trans. Fuzzy Syst.* 25 (6) (2017) 1491–1507.
- [27] P. Zhang, G. Liu, W. Gao, J. Song, Multi-label feature selection considering label supplementation, *Pattern Recognit.* 120 (2021) 108137.
- [28] D. Yeung, D. Chen, E. Tsang, J. Lee, X. Wang, On the generalization of fuzzy rough sets, *IEEE Trans. Fuzzy Syst.* 13 (3) (2005) 343–361.
- [29] Q. Hu, D. Yu, Z. Xie, Information-preserving hybrid data reduction based on fuzzy-rough techniques, *Pattern Recognit. Lett.* 27 (5) (2006) 414–423.
- [30] C. Xiong, W. Qian, Y. Wang, J. Huang, Feature selection based on label distribution and fuzzy mutual information, *Inform. Sci.* 574 (2021) 297–319.
- [31] J. Wan, H. Chen, T. Li, Z. Yuan, J. Liu, W. Huang, Interactive and complementary feature selection via fuzzy multigranularity uncertainty measures, *IEEE Trans. Cybern.* (2023).
- [32] R.E. Schapire, Y. Singer, Boostexter: A boosting-based system for text categorization, *Mach. Learn.* 39 (2) (2000) 135–168.
- [33] P. Zhang, G. Liu, W. Gao, Distinguishing two types of labels for multi-label feature selection, *Pattern Recognit.* 95 (2019) 72–82.
- [34] Y. Zhang, Z.-H. Zhou, Multilabel dimensionality reduction via dependence maximization, *ACM Trans. Knowl. Discov. Data (TKDD)* 4 (3) (2010) 1–21.
- [35] A. Hashemi, M.B. Dowlatshahi, H. Nezamabadi-Pour, Mfs-mcdm: Multi-label feature selection using multi-criteria decision making, *Knowl.-Based Syst.* 206 (2020) 106365.
- [36] A. Hashemi, M.B. Dowlatshahi, H. Nezamabadi-pour, Vmfs: a vikor-based multi-target feature selection, *Expert Syst. Appl.* 182 (2021) 115224.
- [37] M.-L. Zhang, Z.-H. Zhou, Ml-knn: A lazy learning approach to multi-label learning, *Pattern Recognit.* 40 (7) (2007) 2038–2048.
- [38] S. Anwar, N. Barnes, Real image denoising with feature attention, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 3155–3164.
- [39] M. Friedman, A comparison of alternative tests of significance for the problem of m rankings, *Ann. Math. Stat.* 11 (1) (1940) 86–92.

- [40] O.J. Dunn, Multiple comparisons among means, *J. Amer. Statist. Assoc.* 56 (293) (1961) 52–64.

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